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LegalAI Research in LLM Era: Data, Modeling and Evaluation

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Abstract
Legal artificial intelligence (LegalAI) refers to the use of artificial intelligence technologies to automate various legal tasks. Recent advances in large-scale language models have significantly enhanced the capabilities of LegalAI, marking a new stage in its development. In this paper, we present a comprehensive survey of how large language models (LLMs) are reshaping the research paradigm of LegalAI. Beyond improving task performance, LLMs now serve as integral components across the perspectives of data, modeling, and evaluation. We propose a role-based schema that categorizes the involvement of LLMs along these perspectives and use it to systematically review existing studies in three major legal tasks, including legal classification, legal retrieval, and legal generation. Besides, we conduct a detailed quantitative comparison of LLM effectiveness across roles and tasks, and our findings reveal that the impact of LLMs is shaped by both their assigned roles and the nature of the legal tasks.

Keywords Legal artificial intelligence · Large language models · NLP · Literature review

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1 Introduction

Legal artificial intelligence (LegalAI), which refers to the application of AI technologies to automate legal tasks, support decision-making, and enhance the efficiency of legal processes, has become an increasingly prominent focus of academic research. This demand has drawn increasing academic attention to LegalAI, which is now recognized as a critical research direction in both the legal and AI domains (Zhong et al. 2020; Surden 2018). Since most legal information is presented in natural language, LegalAI relies heavily on natural language processing (NLP) techniques. From an NLP perspective, LegalAI research focuses on three core tasks: legal classification, legal retrieval, and legal generation. Recent advances in large language models (LLMs) have substantially enhanced semantic representation, context-aware reasoning, and cross-task generalization. These capabilities have driven widespread adoption of LLMs in LegalAI, not only in practical application but also in academic research, where they are advancing both performance and methodological paradigms. As shown in Fig. 1, the proportion of LLM-related papers in LegalAI publications at conferences and journals has surged dramatically since 2023, rising from 14.9% in 2023 to 42.1% in 2024, and reaching 63.3% in the first half of 2025. As a result, systematically examining how LLMs are transforming the research paradigm of LegalAI is essential for understanding current trends and guiding future research. Existing surveys in LegalAI have primarily focused on specific NLP techniques or individual legal tasks (Zhong et al. 2020; Krasadakis et al. 2024; Cui et al. 2023a; Feng et al. 2024bb; Yang et al. 2024). Although a few recent studies have begun to analyze LLM applications in legal scenarios (Siino et al. 2025; Sun 2023), their broader impact on LegalAI research remains insufficiently explored.

In this paper, we aim to address this gap by providing a comprehensive survey on how LLMs are reshaping the research paradigm of LegalAI. As illustrated in Fig. 2, LLMs are involved throughout the LegalAI research pipeline, including data, modeling, and evaluation. At the data level, they assist in processing both samples and labels to support the creation of LLM-ready datasets. At the modeling level, LLMs enhance method capabilities,

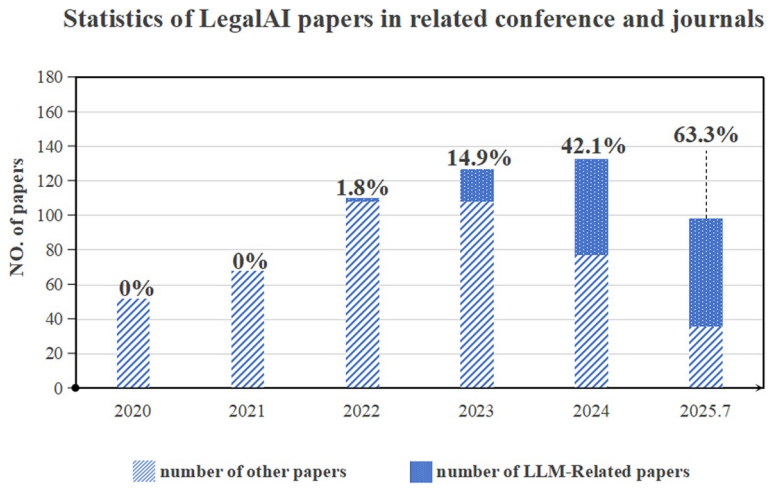


Fig. 1 The proportion of LLM-related papers among LegalAI publications in major conferences and journals, including ACL, AAAI, NeurIPS, ICML, IJCAI, EMNLP, NAACL, COLING, JURIX, information processing & management, artificial intelligence, and artificial intelligence and law

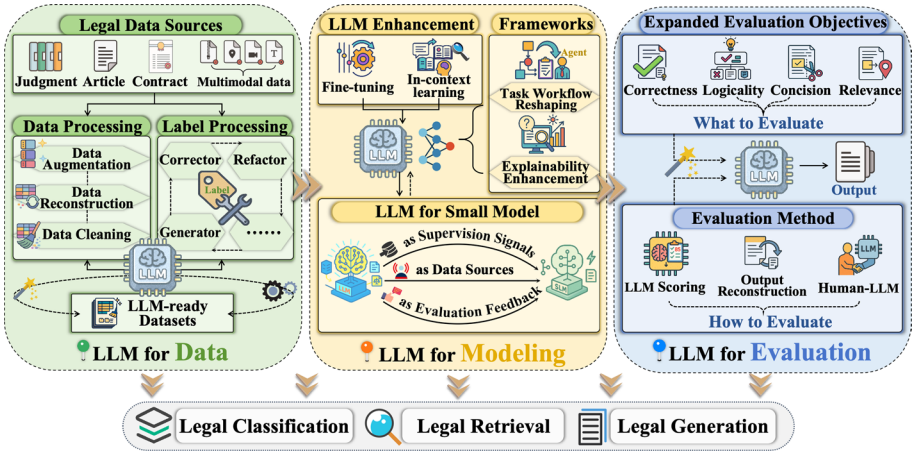


Fig. 2 The roles of LLMs in reshaping LegalAI research

support framework restructuring, and facilitate collaboration among models. At the evaluation level, they broaden assessment objectives and introduce new evaluation methods. The diverse roles of LLMs reflect a shift from a task-centered paradigm that focuses on adapting models to specific tasks toward a model-centered and reasoning-driven paradigm. In this new context, models are no longer limited to solving individual tasks. This integration promotes more structured and coordinated development in LegalAI research.

The key contributions of this survey can be summarized as follows:

- We provide a systematic role-based schema that characterizes how LLMs participate in LegalAI research across data, modeling, and evaluation, establishing a conceptual perspective for understanding the paradigm shift driven by LLMs.
- Building on this schema, we conduct a comprehensive review of existing studies across legal classification, retrieval, and generation tasks, showing how different LLM roles have driven methodological transformations and contributed to the ongoing paradigm shift.
- We conduct a quantitative analysis of LLM contributions from two complementary perspectives: comparing roles within individual tasks and comparing tasks within each methodological role. Our findings highlight that the effectiveness of LLMs is influenced by both their assigned role and the nature of the legal task.
- We construct a comprehensive LegalAI repository that includes papers and datasets involving LLMs.[1]

This paper is organized as follows. Section 2 introduces three research paradigms of LegalAI in the deep learning era. Section 3 introduces a role-based schema to understand the paradigm shift brought by LLMs, by revisiting key challenges in traditional NLP approaches and illustrating the diverse functional roles that LLMs play across LegalAI research. Section 4 categorizes and analyzes existing LegalAI research based on the proposed role-based schema, focusing on three core tasks including legal classification, legal retrieval, and legal

[1] https://github.com/ZJULegalAI/LegalAI-Research-with-LLMs.git.

generation, and introduces several emerging legal scenarios in the LLM era. Section 5 evaluates the effectiveness of LLMs in legal classification, legal retrieval, and legal generation through task performance analysis. Section 6 presents a discussion of the findings across previous sections and discusses future work. Section 7 concludes the paper.

2 Research paradigms of LegalAI in the deep learning era

To better understand the evolution of LegalAI research paradigms in NLP, we highlight three major paradigms that reflect distinct methodological shifts throughout the deep learning era, as shown in Fig. 3.

The first research paradigm of LegalAI, which emerged in the mid-2010 s, marked a transition from statistical learning methods based on handcrafted features to end-to-end neural network modeling. This shift represented a methodological transformation from shallow, task-specific models to deeper semantic representations, driven by advances in deep learning within the field of NLP (Schmidhuber 2015). In this paradigm, LegalAI research focused on task-specific neural representation learning. Each legal task was typically associated with a customized network architecture and training objective. Techniques such as attention mechanisms and multi-task learning were incorporated to enhance semantic extraction and improve the model's capacity to understand complex legal language. Compared with statistical learning, this stage significantly improved semantic modeling capabilities and reduced reliance on manually designed rules, contributing to better accuracy and higher levels of automation. As a result, architectures such as CNNs (LeCun et al. 1989), RNNs (Elman 1990), and LSTMs (Graves 2012) were increasingly applied to core legal tasks including legal text classification (Wei et al. 2018; Undavia et al. 2018; Nejadgholi et al. 2017), legal retrieval (Nanda et al. 2017; Tran et al. 2019), and legal question answering (Kim et al. 2015; Do et al. 2017). However, due to their strong dependence on task-specific contexts, these neural models often lacked generalizability and cross-task transferability.

The second research paradigm of LegalAI began to emerge after 2018, marked by the release of BERT (Devlin et al. 2019), a widely adopted pretrained language model (PLM). This paradigm marked a methodological shift from task-specific neural architectures to transferable semantic representations, as the research focus gradually moved from CNN- and RNN-based models to PLM-based methods. Compared with the first paradigm, the PLM-based approach greatly reduced reliance on handcrafted features and manual architectural design. It enhanced contextual understanding, semantic abstraction, and cross-task generalization, particularly in low-resource legal domains. PLMs such as BERT (Devlin et al. 2019) and RoBERTa (Liu et al. 2019) have been widely applied to core legal tasks, including legal text classification (Chalkidis et al. 2019; Shaheen et al. 2020; Limsopatham 2021), legal retrieval (Rossi and Kanoulas 2019; Shao et al. 2020), and legal question answering (Chen et al. 2023; Wang and Luo 2021), either directly or through domain-adapted variants like Legal-BERT (Chalkidis et al. 2020). These models were initially pretrained on gen-

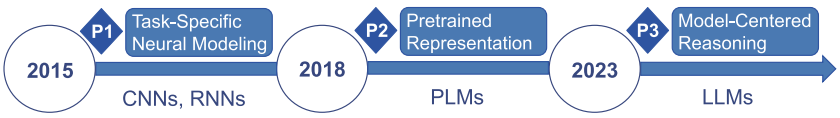


Fig. 3 Three research paradigms of LegalAI in the deep learning era

eral or legal-domain corpora and subsequently fine-tuned for downstream tasks, eliminating the need to redesign model architectures for each individual application. Although PLMs brought substantial advances in semantic representation and facilitated more unified modeling approaches and downstream integration in LegalAI, they continued to rely on static input–output mechanisms. They lacked capabilities for procedural reasoning and handling complex task workflows. These limitations highlighted the need for a subsequent paradigm focused on generative modeling and higher-level reasoning mechanisms.

After 2023, rapid development in large-scale pretrained language models based on the Transformer architecture significantly broadened the scope and effectiveness of LegalAI, marking its third research paradigm, in which the dominant research methodology has gradually shifted toward LLM-based approaches (Siino et al. 2025). Unlike the previous paradigm, which focused on semantic representation, this stage reflects a shift from representation learning to model-centered reasoning. In this paradigm, models not only perform tasks but also restructure the design and resolution strategies of those tasks. General-purpose models such as GPT-4 (Achiam et al. 2023) and Claude, along with legal-specific LLMs like ChatLaw (Cui et al. 2023b) and wisdomInterrogatory (Wu et al. 2026), exhibit strong capabilities in zero-shot generalization, contextual reasoning, multi-turn interaction, and role-based simulation. By reducing reliance on task-specific architectures and extensive annotated data, these models have enabled more scalable and adaptable solutions, transforming legal tasks from static, single-step problem solving into dynamic, multi-step, and interactive workflows. As a result, complex tasks such as legal article interpretation (Jiang and Yang 2023; Wu et al. 2023), case matching (Shu et al. 2024), legal text generation (Su et al. 2025; Liu et al. 2024a), and interpretable judgment prediction (Wei et al. 2025) are no longer addressed in isolation. Instead, they are integrated into unified procedural pipelines involving multi-step prompting, modular reasoning stages, and hybrid collaboration between LLMs and external components. Compared to the PLM stage, LLMs substantially reduce the need for task-specific model design and labeled data, and demonstrate potential for a unified framework that integrates data, modeling, and evaluation.

3 A role-based schema for understanding the LLM paradigm shift

The emergence of LLMs is reshaping the paradigm in LegalAI research. To systematically capture the diverse roles of LLMs in LegalAI research, we propose a three-layer schema consisting of "Perspective", "Object", and "Role", structured along the LegalAI research pipeline, which covers data, modeling, and evaluation (as shown in Table 1). In this schema, perspective refers to the stage of the research pipeline where LLMs are involved; object refers to specific aspects or subtasks within each perspective where LLMs are applied; and role describes the concrete function that LLMs perform for each object. Through a hierarchical structure, this schema clearly identifies the functional components and specific roles of LLMs across the entire research pipeline. It offers a structured analytical framework that integrates diverse research efforts into a unified paradigm, enabling systematic classification, comparative analysis, and in-depth evaluation of LLMs' methodological contributions across different components of LegalAI research. The rest of this paper adopts this schema to examine the roles of LLMs from the three perspectives (data, modeling, and evaluation). We now provide a detailed explanation of this schema by revisiting the major challenges

Table 1 A role-based schema of LLMs in LegalAI research

Perspectives	Objects	Roles
Data	Samples	Data augmentation
		Data reconstruction
		Data cleaning
	Labels	Label generation
		Label correction
		Label reconstruction
Modeling	LLMs	Fine-tuning
		In-context learning
	Small models	Data sources
		Supervision signals
		Evaluation feedback
	Frameworks	Task workflow reshaping
		Explainability enhancement
Evaluation	Dimensions	Expanded evaluation objectives
	Methods	LLM scoring
		Human-LLM
		Output reconstruction

faced by traditional NLP approaches and illustrating the diverse functional roles of LLMs, which collectively drive a broader shift in the research paradigm.

3.1 Perspective of data

Traditional NLP faces two major data challenges in the field of LegalAI. Firstly, legal contexts are open-textured in both common law and civil law systems, such as "reasonable" and "significant impact" (Guitton et al. 2024). These terms are inherently vague and ambiguous, and they require interpretation within specific contexts and legal domains to determine their applicability. When dealing with such semantically open legal texts, traditional NLP methods face significant challenges. High-quality annotated legal corpora remain scarce, especially those that capture context-dependent and ambiguous legal expressions. On the one hand, this scarcity limits models' ability to interpret such expressions accurately; on the other hand, it constrains their ability to generalize across ambiguous or context-dependent legal concepts. Secondly, legal knowledge exhibits significant temporal dynamics, as the continuous revision of legal articles and accumulation of case law require models to have dynamic knowledge representation capabilities. However, traditional NLP models were trained on static datasets that were rarely updated after construction, limiting their adaptability to evolving legal scenarios. Updating such models is costly and labor-intensive, and traditional knowledge graphs are slow to respond to legal revisions (Krasadakis et al. 2024; Quevedo et al. 2023).

To address these challenges, LLMs contribute to improving the quality and adaptability of legal datasets. In LegalAI tasks, data is primarily divided into two objects: samples and labels. Samples consist of the original textual data and its variants used for model training and inference, while labels are the annotations corresponding to these samples. Together, they form the foundation for model training and serve as the core elements ensuring the performance and accuracy of LegalAI systems.

The role of LLMs in **samples** primarily involves data augmentation, data cleaning, and data reconstruction:

- *Data augmentation* refers to a set of techniques that generate synthetic training data from existing datasets, aiming to improve the generalization and accuracy of machine learning models. It enables models to perform better on small or non-representative datasets by enhancing their robustness (Mumuni and Mumuni 2022). In the context of NLP, these methods aim to expand limited labeled datasets by introducing variations that preserve semantic meaning while altering linguistic forms. This strategy helps mitigate overfitting, inject prior knowledge, and simulate diverse test scenarios, addressing challenges inherent to deep learning models trained on small or imbalanced data (Shorten et al. 2021).
- *Data cleaning* refers to the process of detecting, correcting, or removing errors and inconsistencies in datasets to ensure data quality, reliability, and usability for downstream tasks. It involves two primary stages: error detection, i.e., identifying anomalies, and error repair, i.e., correcting or removing erroneous data while minimizing unintended side effects. As dirty data can lead to inaccurate analysis and biased models, data cleaning plays a crucial role in improving input quality and ensuring reliable outcomes (Chu et al. 2016).
- *Data reconstruction* refers to techniques for regenerating or refining textual data to address issues such as incompleteness, noise, and inconsistency. By improving structural coherence and restoring missing or degraded information, it enhances model robustness and supports tasks requiring high semantic or structural fidelity.

The role of LLMs in **labels** primarily involves label generation, label correction, and label reconstruction:

- *Label generation* refers to the automatic assignment of target labels to previously unlabeled data. It aims to reduce reliance on manual annotation, thereby improving data usability and training efficiency.
- *Label correction* refers to the process of identifying and rectifying incorrect or noisy labels within a dataset to improve training data quality and enhance model performance. It focuses on fixing annotation errors caused by human mistakes, ambiguous labeling guidelines, or noise introduced during data collection.
- *Label reconstruction* refers to the process of recovering and reconstructing complete or accurate labels from partially labeled data, noisy annotations, or contextual information. Its core objective is to repair and complete labeling systems, ensuring semantic consistency and task adaptability.

3.2 Perspective of modeling

Traditional NLP methods in LegalAI face three significant modeling challenges. First, legal documents have complex internal structures, often comprising hierarchically nested elements such as articles, sections, and supplementary articles. They also frequently cite other articles or historical versions (Krasadakis et al. 2024; Siino et al. 2025). As a result, accurately interpreting such structures requires hierarchical parsing and contextual reason-

ing, which further increases the difficulty of extracting deep semantic relationships using traditional NLP methods. Secondly, legal reasoning is rule-driven and involves multi-step inference. Specifically, unlike the relatively flexible logical inference in general NLP tasks, legal reasoning must adhere to explicitly defined legal rules and norms. Additionally, legal language is highly formalized, precise in wording, and structurally clear, which facilitates logical reasoning but also increases the difficulty of interpretation (Zhong et al. 2020). Meanwhile, legal cases often require multi-hop reasoning, where information must be extracted and integrated from multiple paragraphs, documents, and even heterogeneous knowledge sources (Siino et al. 2025). This process involves constructing a series of progressively refined intermediate conclusions, which are then used to derive the final judgment through a structured and sequential chain of legal logic. Yet, traditional NLP models often suffer from issues such as skipping steps, missing information, circular reasoning, or spurious correlations when handling such reasoning processes (Zhong et al. 2020; Yu et al. 2024), making them unable to meet the rigor and reliability required for real-world legal reasoning. Lastly, legal reasoning places high demands on both ethical standards and interpretability, requiring a clear, rigorous, and legally coherent reasoning process. However, traditional NLP models primarily focus on pattern recognition and struggle to explicitly represent legal rules, concepts, and reasoning chains (Nazarenko and Wyner 2017). As a result, they are unable to provide causal chains and legal justifications that conform to legal interpretive standards, especially in tasks such as legal judgment prediction and legal question answering. In such scenarios, when a model's output fails to align with legal logic or is based on incorrect facts, the lack of interpretability makes timely identification and correction of errors difficult, potentially leading to legal and ethical risks (Leidner and Plachouras 2017).

LLMs address these modeling-related challenges by enhancing model performance, generalization, and explainability. In LegalAI tasks, modeling involving LLMs can be categorized into three objects: direct enhancement of LLMs, empower small models by integrating LLMs, and innovations at the framework level. Enhancement of **LLMs** focuses on improving their intrinsic capabilities through various techniques, primarily achieved via fine-tuning and in-context learning:

- *Fine-tuning* refers to the process of adapting a pre-trained LLM to a specific legal task or domain by further training it on task-relevant labeled data. This involves updating the model's parameters to better capture domain-specific knowledge, terminology, and reasoning patterns in legal texts. Fine-tuning enables LLMs to specialize and improve their performance on downstream tasks. Compared to training a model from scratch, fine-tuning leverages the general language understanding acquired during pre-training, significantly reducing the required data and computational resources while enhancing accuracy and robustness for targeted legal applications.
- *In-context learning* refers to the capability of an LLM to dynamically adapt to new tasks by utilizing examples provided within the input context, without modifying its internal parameters. The key idea is that the model leverages demonstration examples in a prompt to guide its reasoning and perform the task effectively. For each input instance, the LLM adjusts its output based on external knowledge provided in the input context, enabling flexible and rapid adaptation to a variety of legal scenarios.

Small models (SMs), which are lightweight task-specific models with limited parameters and computational requirements, are empowered by LLMs to improve their performance. It can be divided into three categories: LLMs as data sources, LLMs as supervision signals, and LLMs as evaluation feedback.

- *LLMs as data sources* generate synthetic data based on given instructions and examples to train SMs for specific tasks, especially when labeled data is scarce. This approach reduces costs and accelerates the training process, thereby improving the task performance and generalization capabilities of SMs.
- *LLMs as supervision signals* play a vital role in knowledge distillation, where they guide the training of smaller models by providing soft labels or teacher outputs based on existing data.
- *LLMs as evaluation feedback* play a crucial role by providing task-specific and adaptive assessment metrics. While traditional automated metrics may not always be suitable for specialized tasks, LLMs can automatically evaluate model outputs with greater nuance and contextual understanding.

Innovations at the **framework** level involve structural or architectural changes in LegalAI tasks, where these frameworks coordinate multiple components to tackle complex legal problems more effectively and accurately. LLMs empower frameworks by reshaping task workflows and enhancing explainability.

- *Task workflow reshaping* aims to enhance efficiency, accuracy, and adaptability by redesigning, dynamically optimizing, or automating existing processes. In this context, the optimization focus lies primarily on the task workflow itself, rather than on improving the LLM model during the process.
- *Explainability enhancement* aims to improve the transparency and interpretability of model decisions by generating explanations, visualizing features, and incorporating interactive feedback into workflows.

3.3 Perspective of evaluation

Currently, the evaluation in the field of LegalAI faces two main challenges. On the one hand, the evaluation system is incomplete. Existing metrics mostly focus on traditional accuracy and recall, lacking comprehensive consideration of multiple dimensions such as reasoning completeness, consistency, and emerging issues related to LLMs. This makes it difficult to accurately assess models' legal reasoning abilities and fails to cover new evaluation scenarios arising with the advent of LLMs. On the other hand, the evaluation methods also have limitations. Traditional evaluations often rely heavily on human participation and frequently require reconstruction or manual judgment of model outputs, resulting in low efficiency and high subjectivity, thereby limiting both the accuracy and efficiency of evaluations.

To address these challenges, LLMs contribute by expanding evaluation dimensions and improving evaluation methods. The **evaluation dimensions** focus on the question of what to evaluate, which in turn broadens the evaluation scenarios that have emerged in the LLM era.

- *Expanding evaluation scenarios* includes situations such as assessing models with unlabeled data, broadening the range of evaluation targets, and diversifying evaluation objectives beyond traditional metrics.

The **evaluation methods** focus on how to evaluate, that is, the specific technical means used to carry out evaluation, including approaches such as LLM-based voting and synthetic data evaluation. Evaluation methods can be categorized into three types: LLM scoring, Human-LLM collaboration, and Output reconstruction.

- *LLM scoring* employs LLMs to automatically assess generated outputs by scoring their quality, validity, and relevance based on the model's internal capabilities. This reduces human workload while improving efficiency and consistency.
- *Human-LLM* collaboration combines human judgment with LLM assistance, where human evaluators provide subjective evaluation and guidance, and LLMs offer structured analysis and auxiliary judgments, enabling more comprehensive assessments.
- *Output reconstruction* involves regenerating or verifying outputs to check for logical consistency and completeness, ensuring that model outputs align with expected semantics and legal reasoning, thereby enhancing the accuracy and rigor of evaluation.

4 LLMs in existing LegalAI research

To systematically analyze how LLMs[2] are applied in LegalAI research and how they reflect shifts in research paradigms, this section adopts the proposed role-based schema to review and classify existing studies. We organize current research around three representative legal tasks, namely legal classification, legal retrieval, and legal generation, and investigate the concrete roles played by LLMs in each task scenario, providing grounded support for understanding the methodological transformation driven by LLMs.

4.1 Legal classification

Legal classification is the task of automatically assigning one or more predefined legal categories to texts such as legal articles (Chalkidis et al. 2021a; Liga and Robaldo 2023; Savelka and Ashley 2023), court decisions (Chalkidis et al. 2021b; Xiao et al. 2018; Chalkidis et al. 2019; Spaeth et al. 2024; Nigam et al. 2025, contracts Tuggener et al. (2020); Hendrycks et al. (2021); Lippi et al. (2019)), and other legal documents (Wan et al. 2019; Huber-Fliflet et al. 2024). Its core goal is to enable scalable and fine-grained understanding of legal texts by mapping them into structured legal knowledge through NLP-based classification methods (Cui et al. 2023a; Krasadakis et al. 2024). This not only facilitates efficient access to legal knowledge (Savelka and Ashley 2023; Sargeant et al. 2025), but also generates explicit semantic representations. These representations serve as essential inputs for downstream reasoning and decision-making systems (Wu et al. 2024; Saias and Quaresma 2003; Dragoni et al. 2016).

[2] Unless otherwise specified, LLMs are general-purpose; if an LLM is domain-specific, it will be explicitly indicated.

From a technical perspective, legal classification is a text classification task where the input is a legal text and the output consists of one or more labels denoting legal categories, as shown in Fig. 4. In this architecture, the input token sequence $W = \{w^1, w^2, ..., w^n\}$ is first encoded into contextualized representations $H = \{h^1, h^2, ..., h^n\}$ by an encoder (e.g., LegalBERT or CrimeBERT). The resulting representations are then fed into a classifier, which generates multi-label predictions over the predefined legal label space $L = \{l_1, l_2, ..., l_k\}$ through a sigmoid activation function. Each output dimension corresponds to the probability of a specific legal label l_i.

Legal classification mainly falls into the following four categories. **Legal judgment prediction** (LJP) takes case fact descriptions, evidence texts, and other case related materials as inputs, and outputs judgment outcomes, including legal articles, charges, and prison terms, aiming to assist judicial decision-making and improve court efficiency (Feng et al. 2024aa; Jiang and Yang 2023; Vats et al. 2023; Meng et al. 2025; Huang et al. 2025; Wu et al. 2023; Li et al. 2025). **Legal topic classification** takes legal texts as input and assigns labels representing the subject matter, thematic direction, or issue category discussed in the text, facilitating efficient organization, retrieval, and analysis of legal information (Prasad et al. 2024; Sargeant et al. 2025; Huber-Fliflet et al. 2024; Chen et al. 2024a, b; Drápal et al. 2023; Jung and Jung 2025). **Compliance and applicability classification** takes legal articles and behavioral descriptions as input, and outputs judgments on whether the articles apply and whether the conduct complies, thereby assisting enterprises automatically identify compliance risks (Fan et al. 2024; Hassani 2024). **Legal structural classification** takes sentences or paragraphs within legal documents as input and assigns functional roles that reflect each text segment's position within the overall document structure, enabling models to represent the internal organization of legal texts, the semantic roles of articles or clauses, and the interpretive components of legal reasoning (Liga and Robaldo 2023; Šavelka and Ashley 2018; Belfathi et al. 2023; Savelka and Ashley 2023; Gray et al. 2024). This classification supports a wide range of downstream applications, including legal summarization (Kanapala et al. 2019), argument mining (Lawrence and Reed 2019), and fact-based reasoning (Nigam and Deroy 2023).

LLMs are involved throughout the entire process of legal classification tasks, supporting innovations in system design and improvements in classification accuracy. In the following, we provide a structured analysis of how LLMs contribute to LegalAI research across

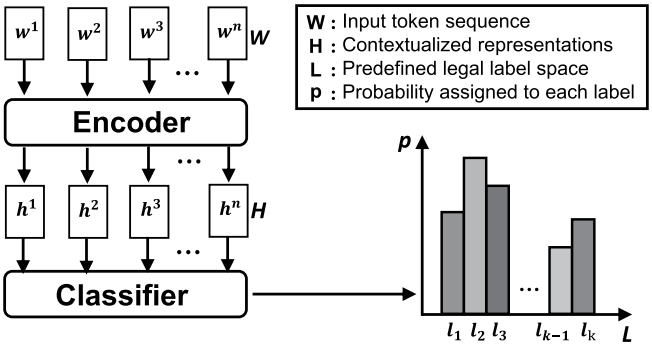


Fig. 4 The pipeline of multi-label legal classification tasks

various classification scenarios. As shown in Fig. 5, we organize the roles of LLMs in legal classification tasks according to the schema introduced in Table 1.

4.1.1 LLM for data

We analyze how LLMs address data challenges across both sample and label objects.

Sample Compared to general classification tasks, legal classification relies more heavily on precise semantic cues and consistent structural patterns to distinguish between closely related labels. It requires input texts that are semantically accurate, structurally consistent, and contextually coherent. Even minor imperfections in the input such as incompleteness, inconsistency, or noise can obscure semantic boundaries, disrupt structural patterns, and introduce training errors, ultimately interfering with label recognition and reducing classification accuracy. Furthermore, the increasing variety of classification tasks calls for more diverse datasets across legal domains and document structures at the sample level. LLMs, with their strong capabilities in contextual understanding, text generation, and task alignment, can enhance data diversity, reconstruct incomplete samples, and clean low-quality inputs, while ensuring that the generated or processed texts are semantically accurate, structurally coherent, and aligned with the objectives of legal classification tasks. In the following, we analyze how LLMs contribute specifically via data augmentation, reconstruction, and cleaning.

In the role of **data augmentation**, GOLDCOIN automatically generates synthetic privacy cases that incorporate roles, information, and transmission principles, based on Contextual Integrity Theory and the strong semantic alignment and structural generation capabilities

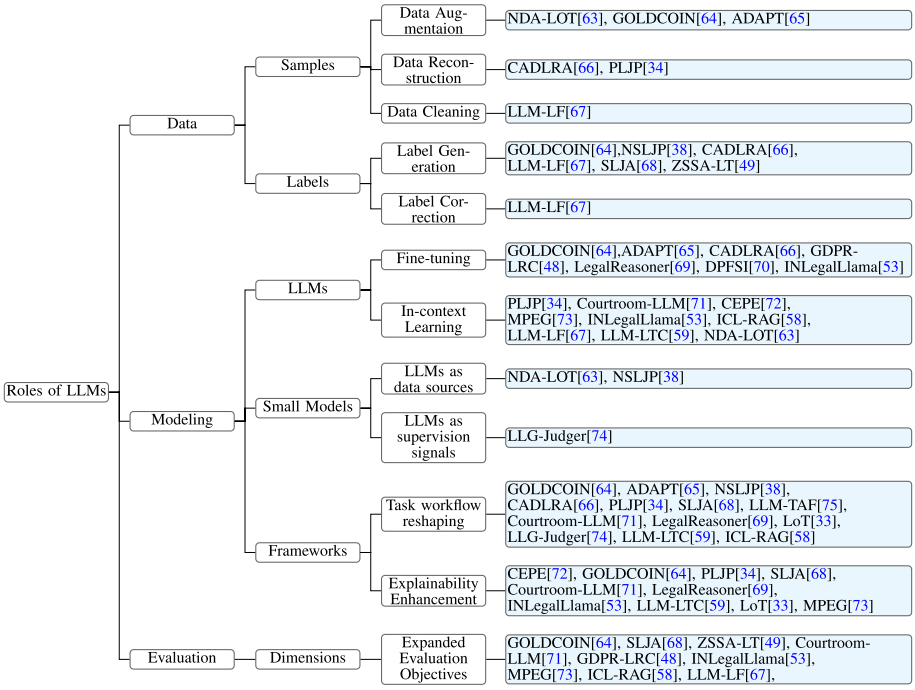


Fig. 5 Roles of LLMs in legal classification tasks

of LLMs. The augmented data are subsequently evaluated through automatic filtering and human expert review, with consistency, integrity, and diversity checks confirming their legal validity and quality (Fan et al. 2024). NDA-LOT constructs large-scale synthetic texts representing overruling cases based on pre-trained language models and prompt engineering, aiming to address data scarcity in legal corpora. By combining the generative capabilities of GPT models with the linguistic patterns of legal precedents, NDA-LOT automatically generates training samples with overruling semantics, significantly improving data coverage and the model's ability to distinguish between binary labels (Sheik et al. 2024). ADAPT constructs multi-task synthetic reasoning trajectories using LLMs, covering all reasoning steps from fact element extraction to final judgment prediction. These trajectories provide structured and logically consistent training data for fine-tuning smaller LLM models, effectively addressing the lack of supervision on intermediate legal reasoning processes (Deng et al. 2024a).

In the role of **data reconstruction**, CADLRA utilizes LLMs to reconstruct fact descriptions into structured representations of criminal behavior. This process corrects noisy or unstructured inputs and produces semantically aligned, structurally consistent representations, thereby enhancing charge prediction under complex and ambiguous fact conditions (Feng et al. 2024aa). PLJP employs GPT−3.5 to reorganize fact descriptions into three components, including subjective motivation, objective behavior, and ex post facto circumstances. This reorganization improves the semantic clarity and logical coherence of the input, facilitates the retrieval of relevant precedents, and provides the model with more structured representations, thereby enhancing reasoning performance in multi-label prediction tasks (Wu et al. 2023).

In the role of **data cleaning**, LLM-LF integrates automated preprocessing with instruction-driven generation using LLMs to address common issues in court judgments. It filters out non-informative segments, extracts semantically relevant information, standardizes input structures, and refines candidate legal factors. These steps enhance the structural and semantic consistency of the data, thereby improving the accuracy and robustness of downstream classification models (Gray et al. 2024).

Label Legal classification tasks are inherently label-driven and often involve complex, multi-dimensional annotation schemes. Legal texts must be assigned to labels such as legal articles, case types, normative functions, or rhetorical roles. These label taxonomies are typically multi-label, hierarchical, and sometimes semantically ambiguous, where a single text may simultaneously reflect multiple legal perspectives. Since classification models rely on these labels as supervision signals during training, the quality and structure of the label space are critical to overall task performance. This complexity places high demands on the quality, consistency, and completeness of label annotations. To address these challenges, LLMs, trained on legal corpora and equipped with strong reasoning and instruction-following abilities, can provide effective support for improving the structure and reliability of label annotations. Specifically, LLMs can assist in automatically generating structured legal labels, correcting ambiguous or inconsistent annotations, and reconstructing incomplete label sets, particularly in hierarchical multi-label settings. These capabilities contribute to better alignment between texts and labels, improve the robustness of label annotations, and ultimately support more accurate and generalizable legal classification models. In the following, we analyze how LLMs contribute via label generation and correction specifically.

In the role of **label generation**, GOLDCOIN employs LLMs to simultaneously generate case backgrounds and their corresponding legal conclusions (e.g., Applicable and Not Applicable for applicability tasks, and Permit and forbid for compliance tasks) based on contextual integrity theory (Fan et al. 2024). CADLRA employs LLMs to transform traditional label selection into generative charge labeling. The method combines legal article retrieval with prompt-based generation, guiding LLMs to dynamically generate charge labels based on fact description and retrieved legal articles (Feng et al. 2024aa). NSLJP utilizes LLMs to generate fact elements as potential labels. Specifically, LLMs extract key variables that serve as logical variables in first-order logic rules to infer structured judgment outcomes (Wei et al. 2025). ZSSA-LT utilizes GPT models to perform zero-shot label generation in legal classification tasks. By providing concise type definitions, the model automatically generates semantic labels for adjudicatory decisions, contract clauses, and legal articles. This approach eliminates the need for manual annotations and achieves performance comparable to SMs across multiple tasks (Savelka and Ashley 2023). SLJA utilizes LLMs to generate intermediate legal labels such as criminal elements and article interpretations to support multi-stage judgment reasoning. It uses prompt-based generation with ChatGPT to produce structured legal representations that support final outcome prediction (Deng et al. 2023). LLM-LF uses LLMs to extract legally relevant fact elements from case texts as implicit labels. Based on instruction-driven prompting with LLMs, the model automates the identification of these elements, replacing manual rule-based annotation with context-aware label generation (Gray et al. 2024).

In the role of **label correction**, LLM-LF refines LLM-generated legal factors by combining identical or highly similar induced factors, clarifying definitions, and correcting outputs that are overly broad or not legally meaningful. This refinement process, conducted by either humans or LLMs using structured prompts with LLMs, serves as a form of label correction that improves the consistency and legal relevance of the resulting factor representations (Gray et al. 2024).

4.1.2 LLM for modeling

We analyze how LLMs contribute to legal classification modeling across three objects, namely LLMs, small models, and frameworks.

LLMs Compared to general text classification tasks, legal classification requires models to handle normative, ambiguous, and context-sensitive labels within structurally complex label systems that are often hierarchical and semantically overlapping. These tasks involve multi-level semantic dependencies, long-tail label distributions, and complex reasoning across lengthy documents, making adaptability and precise semantic modeling essential. LLMs offer a promising solution due to their strengths in semantic generalization, long-context reasoning, and the ability to model hierarchical and correlated label structures. Fine-tuning enables LLMs to internalize nuanced legal semantics, capture domain-specific labeling criteria, and learn stable decision boundaries by adapting pretrained models with annotated legal datasets, which is particularly effective for hierarchically dependent labels and evolving legal norms. Meanwhile, in-context learning allows LLMs to dynamically adjust their interpretation and prediction strategies based on case-specific context, improving classification accuracy without additional parameter updates and reducing reliance on

extensive labeled data. In the following, we specifically analyze fine-tuning and in-context learning.

In the role of **fine-tuning**, GOLDCOIN fine-tunes four instruction-tuned LLMs via parameter-efficient LoRA adaptation on synthetic cases grounded in Contextual Integrity theory (Fan et al. 2024). ADAPT enhances LLMs through fine-tuning on multi-task synthetic trajectories, enabling the model to internalize complex semantic dependencies and hierarchical label structures inherent in LJP tasks. This fine-tuning strategy addresses the challenge of distinguishing between confusing charges and improves performance on multi-label classification tasks involving complex cases and similar charges (Deng et al. 2024a). CADLRA fine-tunes the pre-trained ChatGLM using techniques such as P-Tuning v2 Parameter Effective Fine-Tuning, enabling the model to better adapt to the input–output format of multi-charge prediction tasks (Feng et al. 2024aa). GDPR-LRC fine-tunes GPT-3 on a dataset of GDPR legal articles annotated with LegalDocML and LegalRuleML, improving classification accuracy across multiple legal rules by capturing subtle semantic and structural differences between rule types (Liga and Robaldo 2023). LegalReasoner fine-tunes LLMs on their legal corpus to improve overall performance in LJP tasks (Wang et al. 2024). DPFSI is fine-tuned with a deontic logic prompt to improve the semantic encoding of deontic expressions in legal texts (Meng et al. 2025). INLegalLlama fine-tuned LLMs on the PredEx training dataset, which contains Indian legal court decisions and expert-annotated explanations. This supervised fine-tuning bridges the gap between the model's general knowledge and the specialized demands of legal reasoning tasks, enhancing its reasoning abilities and capacity to effectively comprehend and apply legal principles (Nigam et al. 2025).

In the role of **In-context learning**, PLJP incorporates relevant precedents to enable LLMs to dynamically understand and compare differences between a case and the relevant precedents. This precedent-based contextual understanding facilitates more accurate predictions by effectively leveraging contextual information without modifying model parameters (Wu et al. 2023). Courtroom-LLM integrates similar-case examples as contextual demonstrations, enabling Judge-LLMs to infer labels through analogy and deliberation. This precedent-based prompting allows the model to perform classification dynamically within context (Jung and Jung 2025). CEPE leverages the in-context learning ability of LLMs by combining few-shot examples with Chain-of-Thought reasoning prompts, allowing the model to dynamically adapt to the task of classifying rhetorical roles in legal judgment texts (Belfathi et al. 2023). MPEG designs a prompt for in-context training in a few-shot setup. The prompt explicitly provides examples of case facts and relevant legal articles to guide the model in learning the correspondence between facts and articles. Within the few-shot contextual examples, the LLM dynamically adjusts its prediction strategy, generating a list of the most applicable articles along with explanations based on the fact statements (Vats et al. 2023). ICL-RAG designs structured few-shot prompts to guide the LLM in identifying delay-related statements in legal documents. Without updating model parameters, it leverages retrieval-augmented generation (RAG) to retrieve the most similar labeled examples as contextual information, significantly improving classification accuracy in legal document review tasks (Huber-Fliflet et al. 2024). LLM-LF infers the structure and semantics of legal factors from a task description and a cross-domain example through few-shot prompting (Gray et al. 2024). INLegalLLaMA employs structured prompts containing exemplar case–decision pairs, enabling LLMs to infer legal reasoning patterns and generate both predic-

tions and explanations based on contextual understanding (Nigam et al. 2025). LLM-LTC employs a contextual few-shot in-context learning approach, in which the LLM is guided by additional contextual information relevant to topic modelling and a predefined list of legal topics from which it must select the most appropriate one (Sargeant et al. 2025). NDA-LOT utilizes few-shot prompting to leverage the in-context learning capabilities of GPT models by providing a small subset of labeled legal overruling data as examples, guiding LLMs to generate synthetic labeled samples for the legal overruling classification task (Sheik et al. 2024).

Small Models With sufficient training data, legal SMs can perform classification tasks efficiently and at low cost. However, in real-world scenarios, limited data quantity and quality hinder their ability to learn effective representations of low-frequency labels and to capture complex semantic relationships between normative structures and factual reasoning. Their limited parameter capacity and reliance on hard labels further restrict the model's ability to learn probabilistic label distributions and semantic overlaps across classes. LLMs address these challenges by generating representative samples for underrepresented classes, providing soft labels, and offering pretrained semantic features that guide training and enhance semantic understanding. In the following, we analyze the use of LLMs as data sources and supervision signals for SMs.

As **data sources**, NDA-LOT synthesizes training samples for the legal overruling task using GPT models, which augments the training data for small deep learning models and improving their classification performance (Sheik et al. 2024). NSLJP generates structured fact elements from civil case texts using GPT-4, thereby providing labeled inputs that guide the training of the small neural network module within its end-to-end framework (Wei et al. 2025). As **supervision signals**, LLG-Judger integrates ChatGPT's predictions as soft supervision signals via KL divergence, thereby guiding the training of the GNN module and enhancing its semantic discrimination in LJP tasks (Xia and Luo 2024).

Frameworks Compared to general classification tasks, legal classification involves complex workflows with multi-step reasoning, hierarchical label structures, and context-dependent legal semantics. Traditional frameworks often treat legal classification as a static, one-shot process, lacking both flexibility and interpretability. They typically produce output labels based on hidden semantic representations learned from training data, without revealing the underlying reasoning behind their decisions. This makes it difficult to understand how the model distinguishes between closely related categories, especially when labels share similar legal language, involve overlapping legal articles, or rely heavily on contextual interpretation. LLM-based frameworks address these issues by reshaping the task workflow through dynamically decomposing the process into subtasks, integrating reasoning chains, and embedding natural language explanations. In the following, we specifically analyze two roles: task workflow reshaping and explainability enhancement.

In the role of **task workflow reshaping**, GOLDCOIN restructures legal classification into a multi-step reasoning process using GPT-4, guiding LLMs to perform contextual feature extraction, norm retrieval, and legal judgment in sequence (Fan et al. 2024). ADAPT decomposes LJP into a multi-step workflow: extracting key elements, using them to distinguish between similar charges, and making the final decision based on the reasoning from earlier steps. This enables LLMs to better follow human-like legal reasoning compared to traditional one-shot classification (Deng et al. 2024a). NSLJP reshapes the LJP workflow into a structured fact-based workflow by using GPT-4 to extract key factual elements

from civil cases, enabling logic-based judgment through rule matching rather than direct label prediction (Wei et al. 2025). CADLRA restructures charge prediction by transforming it from a static multi-label classification task into a multi-stage workflow involving criminal act extraction, law article retrieval, and charge generation using LLMs (Feng et al. 2024a). PLJP leverages LLMs to restructure LJP into a multi-step process, where the LLM sequentially analyzes fact descriptions, extracts legal elements, compares them with precedents, and produces final decisions (Wu et al. 2023). SLJA formalizes LJP as a four-step syllogistic reasoning workflow, comprising article retrieval, criminal element generation, article interpretation, and final judgment, thereby aligning the task execution with structured legal reasoning practices (Deng et al. 2023). LLM-TAF restructures thematic analysis into a multi-step workflow, where the LLM iteratively generates initial codes, incorporates expert feedback to trigger re-generation, and progressively aggregates codes into potential and high-level themes before producing final thematic predictions for legal data points (Drápal et al. 2023). Courtroom-LLM reshapes the text classification workflow into a structured, courtroom-inspired process consisting of a preliminary hearing and a main trial phase, thereby improving interpretability and accuracy in handling ambiguous cases (Jung and Jung 2025). LegalReasoner reshapes the LJP task workflow by introducing a multi-hop reasoning stage implemented through a hierarchical attention transformer, enabling structured legal inference across fact descriptions, articles, and precedents (Wang et al. 2024). LoT reshapes the LJP workflow by prompting LLMs to explicitly follow a three-step deductive reasoning structure, including article identification, fact recognition, and final judgment, thereby transforming the task from one-shot classification into a structured legal reasoning process (Sheik et al. 2024). LLG-Judger guides ChatGPT through step-by-step prompts to generate intermediate fact descriptions and reasonable legal predictions (Xia and Luo 2024). LLM-LTC reshapes the traditional topic modelling workflow by designing a prompt-based classification pipeline driven by LLMs. Through iterative prompt refinement, self-evaluation, and manual validation, the LLM directly assigns legal topics from a predefined taxonomy (Sargeant et al. 2025). ICL-RAG integrates RAG with in-context learning to retrieve the most similar labeled examples and dynamically incorporate them into stnow-Proructured prompts for delay-related text identification (Huber-Fliflet et al. 2024).

In the role of **explainability enhancement**, CEPE improves explainability in rhetorical role prediction by guiding GPT−3.5 to generate step-by-step reasoning and contextual justifications for its label predictions (Belfathi et al. 2023). GOLDCOIN improves explainability by guiding LLMs to produce structured multi-step outputs through Chain-of-Thought prompting. For both applicability and compliance tasks, the model generates intermediate reasoning steps to extract context-specific features, thereby making the classification process more interpretable and transparent (Fan et al. 2024). PLJP improves explainability by prompting LLMs to output predicted legal results along with textual justifications referencing facts and precedents, which clarify the reasoning path behind predictions (Wu et al. 2023). SLJA improves explainability by structuring LJP as a syllogistic reasoning process using GPT models. By explicitly generating the major premise, minor premise, and conclusion, the model illustrates the legal logic behind its predictions and offers detailed interpretations for each decision (Deng et al. 2023). Courtroom-LLM improves explainability by simulating legal roles and organizing multi-LLM deliberation in structured stages. This enables step-by-step reasoning, incorporation of diverse perspectives, and output of interpretable decisions, especially for ambiguous cases (Jung and Jung 2025). LegalReasoner

improves explainability by integrating LLMs into a multi-stage workflow that includes multi-hop reasoning and judgment synthesis. The model generates intermediate outputs such as fact extraction, legal analysis, and precedent comparison, producing coherent and legally grounded explanations (Wang et al. 2024). INLegalLLaMA improves explainability through supervised fine-tuning, enabling LLMs to generate predictions along with comprehensible explanations (Nigam et al. 2025). LLM-LTC improves explainability by prompting LLMs to justify its topic classification through brief reasoning and references to relevant legal principles or precedents, thereby enhancing the transparency and traceability of its decisions (Sargeant et al. 2025). LoT instructs GPT-3 models to generate structured legal reasoning in the form of syllogism, comprising a major premise, a minor premise, and a conclusion. This format enables the model to provide both the decision and its justification (Jiang and Yang 2023). MPEG improves explainability by prompting LLMs to generate natural language rationales alongside its predictions, producing outputs that align with actual court reasoning (Vats et al. 2023).

4.1.3 LLM for evaluation

Compared to general classification tasks, legal classification presents distinct evaluation challenges due to the complexity, ambiguity, and normative nature of legal texts. Traditional evaluation methods, which rely on fixed datasets and static metrics, often fail to capture these characteristics, limiting their ability to assess how well models address legal-specific challenges. LLMs provide deeper insights into model performance by enabling more diverse and realistic evaluation settings that go beyond fixed references. They can generate structured outputs such as labels or factor representations, which are then assessed for their performance, thereby offering a more comprehensive understanding of model behavior under complex semantic and normative demands. However, as current legal classification tasks still rely heavily on fixed benchmark datasets with well-defined output formats, the use of LLMs for evaluation in this domain remains relatively rare. Nevertheless, the emergence of LLMs has introduced some new evaluation dimensions.

To extend **evaluation dimensions**, GOLDCOIN compares direct prompting with multi-step Chain-of-Thought (CoT) prompting and finds that while CoT can improve evaluation performance in some models, its effectiveness depends on whether the model is properly aligned with the legal task (Fan et al. 2024). SLJA compares the performance of LLMs with and without syllogistic prompting to examine how structured legal reasoning impacts model effectiveness on legal judgment tasks (Deng et al. 2023). ZSSA-LT evaluates LLMs from the perspective of cost by comparing the performance and expense of batch prediction versus single-instance prompting. The study shows that while predicting one data point at a time yields a slightly higher F1 score, it incurs more than twelve times the cost, highlighting a practical trade-off between marginal performance gains and economic viability (Savelka and Ashley 2023). Courtroom-LLM compares sequential and parallel deliberation setups against traditional single-LLM classifiers and simpler multi-LLM setups, evaluating how different collaboration structures affect classification performance (Jung and Jung 2025). GDPR-LRC compares the performance of two different prompt strategies across four legal rule classification scenarios to examine how prompt design influences fine-tuned LLM's classification accuracy (Liga and Robaldo 2023). INLegalLLaMA expands the evaluation perspective by assessing the quality of LLM-generated explanations (Nigam et al. 2025).

ICL-RAG compares zero-shot and few-shot prompting to examine how prompt formulation and example selection impact classification performance (Huber-Fliflet et al. 2024). MPEG evaluates the quality of explanations using metrics that capture the semantic alignment between generated and expert-annotated rationales, thereby expanding the evaluation objectives from predictive performance to interpretability and reasoning quality (Vats et al. 2023). LLM-LF compares factor labels optimized by LLMs and human experts to evaluate their impact on legal outcome prediction (Gray et al. 2024).

4.2 Legal retrieval

Legal retrieval, or legal information retrieval, is the task of obtaining relevant legal documents, such as legal articles, case law, and sections of legal literature, from external legal data collections (Kobayashi and Takeda 2000; Singhal et al. 2001). Given the overwhelming volume and inherent complexity of legal information, this task remains highly labor-intensive and constitutes a significant portion of the work of legal professionals (van Noortwijk 2017). However, the field is constantly evolving through emerging technologies and approaches aimed at enhancing retrieval efficiency and accuracy (Locke and Zuccon 2022). This progression is crucial as it facilitates efficient access to relevant legal knowledge and provides reliable information foundations essential for informed legal analysis and decision making.

From a technical perspective, legal retrieval constitutes a document ranking task where the input is a legal query and the output comprises the top-k relevant documents from a legal corpus, as depicted in Fig. 6. In this architecture, the query token sequence W^Q and each token sequence W_i^D in the corpus is encoded through a legal-domain encoder, which may adopt one of three configurations: a Biencoder (separate query/document encoders), a Shared Encoder (shared query/document encoder), or a Cross Encoder (joint query-document encoding). The encoder generates dense vector representations: the query embedding v^Q and a set of document embeddings $\{v_i^D\}_{i=1}^m$. These embeddings are then directly fed into a similarity calculation module (the retriever), which computes relevance scores using a distance metric (typically cosine similarity). The documents are ranked by descending similarity scores, and the system returns the top-k retrieved results with the highest relevance to the query.

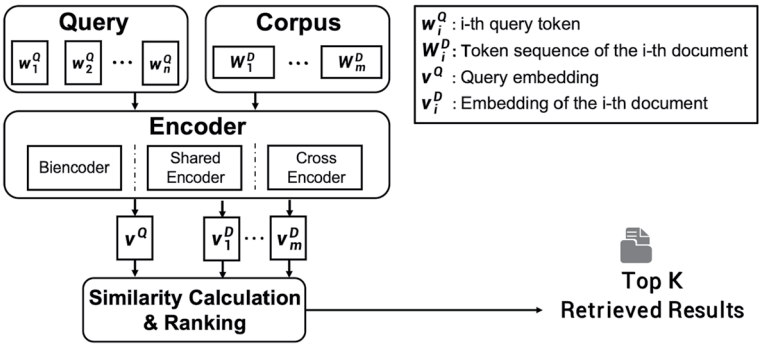


Fig. 6 The pipeline of legal retrieval tasks

Legal retrieval sub-tasks are primarily categorized according to their specific retrieval targets, falling into four distinct types. **Legal article retrieval** focuses on retrieving statutory texts such as legal articles and local ordinances to provide foundational legal authority (Nguyen et al. 2024a, b, c, d, 2025a, b, c; Tjandra et al. 2025; Wehnert et al. 2025; Wu et al. 2025). **Case law retrieval** aims to identify precedent cases that are factually or legally similar to the query case, thereby supporting judicial consistency (Ma et al. 2024; Nguyen et al. 2024d; Deng et al. 2024b; Shu et al. 2024; Nguyen et al. 2024b, 2025b, c; Wehnert et al. 2025; Wu et al. 2025; Tang et al. 2025). **Scholarly works retrieval** accesses academic materials such as articles, books, and commentaries, to facilitate doctrinal analysis and theoretical grounding (Nguyen et al. 2024a). **Litigation document retrieval** encompasses diverse materials including patents, trademarks, company records, and other evidence pertinent to legal disputes, enabling more comprehensive case preparation (Liu et al. 2024b; Jang and Stikkel 2024; Qin et al. 2024). This categorization enables purpose-built retrieval methodologies that directly support critical downstream legal applications like legal article analysis or legal due diligence reporting.

LLMs are involved throughout the entire process of legal retrieval tasks, supporting innovations in system design and improvements in retrieval accuracy. In the following, we provide a structured analysis of LLM contributions to LegalAI research in legal retrieval tasks. As shown in Fig. 7, we organize the roles of LLMs in legal retrieval tasks according to the taxonomy introduced in Table 1.

4.2.1 LLM for data

We analyze how LLMs address data challenges across both sample and label objects.

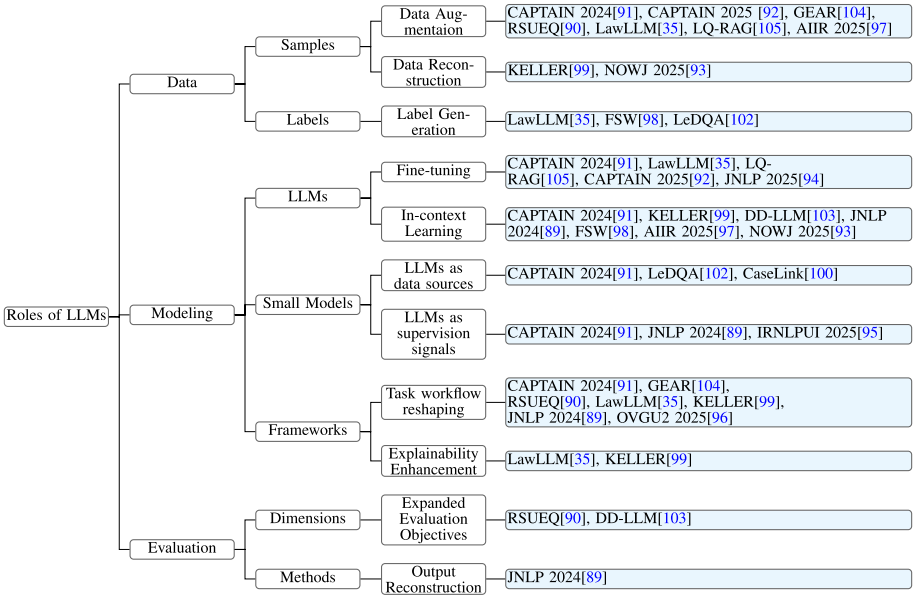


Fig. 7 Roles of LLMs in legal retrieval tasks

Sample Legal retrieval at the sample level requires precise query–document alignment and contextual fidelity across diverse legal domains. The specialized terminology, implicit reasoning chains, and context-dependent interpretations in legal language make it difficult to construct sufficient high-quality training data, while fragmented or poorly structured inputs can obscure semantic relationships and hinder accurate precedent matching. Conventional augmentation and reconstruction methods often fail to preserve domain-specific semantics or restore the logical flow essential for retrieval. LLMs, with strengths in contextual comprehension, structural regeneration, and domain-aware text generation, can synthesize semantically faithful queries and contextually aligned legal texts that maintain factual consistency and legal reasoning structures. This bidirectional generation capability enriches the coverage of query intent variations, reconstructs unstructured data into coherent formats, and enhances retrieval models' capacity for recognizing latent legal concepts, improving cross-jurisdictional generalization, and strengthening relevance matching in complex legal retrieval tasks. In the following, we analyze how LLMs contribute via data augmentation and reconstruction specifically.

In the role of **data augmentation**, RSUEQ applies LLMs for zero-shot legal term extraction and query reformulation, expanding retrieval inputs with implicit legal concepts, and evaluates the augmented queries by comparing recall improvements and performance gains across COLIEE 2022 and 2023 datasets (Nguyen et al. 2024c). LQ-RAG utilizes LLMs to synthesize query-context pairs for embedding model training (Wahidur et al. 2025). GEAR employs the specialized legal LLM ChatLaw to generate statutory-definition-based keywords that strengthen semantic representations for retrieval, and the augmented results are subsequently reviewed by legal experts to ensure their quality and prevent hallucination (Qin et al. 2024). CAPTAIN 2024 uses LLMs to summarize premises and synthesize hypotheses for one of its COLIEE tasks to mitigate data scarcity (Nguyen et al. 2024b). CAPTAIN 2025 utilizes LLMs to perform causal relationship extraction from legal articles, decomposing complex premises into structured cause-effect pairs to generate refined synthetic data that enhances the model's comprehension of legal logic and improves entailment judgment accuracy (Nguyen et al. 2025b). LawLLM generates summarized case facts and verdicts for downstream task training (Shu et al. 2024). AIIR 2025 employs LLMs to augment training data by generating multiple paraphrased, summarized, and simplified variants of queries using ten distinct prompts per sample, thereby expanding positive query-article pairs for retrieval (Wu et al. 2025).

In the role of **data reconstruction**, KELLER employs LLMs to rewrite raw case descriptions into structured legal sub facts. This reconstruction optimizes the alignment between the query intention and document structure, significantly improving precedent retrieval accuracy in complex case-matching scenarios (Deng et al. 2024b). NOWJ 2025 utilizes LLMs to perform abstractive summarization on lengthy legal cases via zero-shot prompting, compressing extensive documents into concise summaries while preserving key factual and logical structures to enhance subsequent relevance comparison (Nguyen et al. 2025a).

Label Legal retrieval at the label level demands precise relevance annotations aligned with evolving legal articles, jurisdiction-specific interpretations, and complex legal taxonomies. Conventional manual annotation is costly and often produces incomplete or inconsistent labels due to subjective interpretations, limiting both scalability and semantic fidelity across diverse jurisdictions. LLMs pre-trained on legal corpora address these challenges through structured semantic comprehension, enabling the automated generation of coherent

and legally grounded annotations that preserve semantic hierarchies, jurisdictional nuances, and task relevance. This capability expands the coverage of complex legal relationships within training data, mitigates issues of annotation sparsity, and enhances retrieval models' capacity to recognize latent precedent connections and to improve cross-system compatibility. In legal retrieval, LLMs are rarely used in label reconstruction and correction because relevance annotations are inherently subjective and lack definitive ground truth, making reliable correction both technically and legally challenging. In the following, we analyze how LLMs contribute via label generation specifically.

In the role of **label generation**, LawLLM utilizes GPT-4 to synthesize categorical verdict labels (e.g., plaintiff wins, defendant wins) from raw case law, thereby transforming unstructured narratives into classification-ready annotations (Shu et al. 2024). FSW employs LLMs to automatically assign multi-grade relevance scores to case descriptions, generating granular retrieval-specific pseudo-labels for training robustness (Ma et al. 2024). LeDQA leverages SOTA LLMs for sentence-level relevance annotation and automated legal element labeling, constructing QA datasets with minimal human intervention to reduce annotation costs while maintaining accuracy (Liu et al. 2024b).

4.2.2 LLM for modeling

We analyze how LLMs contribute to legal retrieval modeling across three objects, namely LLMs, small models, and frameworks.

LLMs Unlike general information retrieval, legal retrieval necessitates models that dynamically interpret laws and cases as they evolve across time and jurisdictions, requiring adaptive reasoning over complex statutory frameworks. LLMs offer distinct advantages for these challenges through their capacity for normative pattern recognition, cross-document inference, and modeling hierarchical and correlated label structures. Fine-tuning enables targeted domain adaptation, preserving general linguistic capabilities while internalizing legal semantics and relationships between queries and corpora without the prohibitive cost of full-parameter training. In-context learning further enhances adaptability by conditioning retrieval on dynamically provided legal exemplars, allowing models to navigate jurisdictional variations, shifting legal article interpretations, and conflicting precedents without architectural changes. The following examines these two enhancement roles in detail.

In the role of **fine-tuning**, CAPTAIN 2024 integrates LoRA with Flan-T5, selectively updating weight matrices to optimize multi-task legal inference without exhaustive retraining, thereby balancing computational economy with jurisdictional accuracy (Nguyen et al. 2024b). LawLLM employs LoRA-based instruction tuning on Gemma-7B, transforming the base model into a unified architecture that internalizes cross-jurisdictional reasoning patterns; this method sustains coherence across diverse retrieval operations, from case law matching to regulatory compliance checks (Shu et al. 2024). LQ-RAG fine-tunes embedding LLMs via Multiple Negatives Ranking Loss using synthetic query-context pairs to improve relevance discrimination in retrieval tasks (Wahidur et al. 2025). CAPTAIN 2025 integrates LoRA with LLM models, employing contrastive learning objectives for candidate retrieval and cross-entropy optimization for entailment verification, thereby enabling efficient multi-model ensemble refinement while preserving domain-specific legal reasoning capabilities (Nguyen et al. 2025b). JNLP 2025 adopts a multi-stage refinement: embedding LLMs undergo fine-tuning via LoRA/QLoRA to mitigate class imbalance in binary rel-

evance classification, while generative LLMs leverage identical parameter-efficient adaptation for contextual legal article comprehension; merging these components produces a hybrid pipeline that simultaneously optimizes recall and enhances competitive F2 performance in cross-jurisdictional retrieval scenarios (Nguyen et al. 2025c).

In the role of **in-context learning**, CAPTAIN 2024 deploys hybrid prompting, zero-shot inference LLMs in COLIEE Task 2, and few-shot demonstrations curated through BM25-retrieved analogous verdicts in Task 4, to adaptively resolve statutory interpretation variances across jurisdictions (Nguyen et al. 2024b). KELLER coordinates crime classification and sub-fact summarization entirely through task-specific prompting chains, eliminating the need for parametric adjustments while maintaining procedural rigor (Deng et al. 2024b). DD-LLM leverages GPT-4 for few-shot relevance identification in due diligence contexts, countering class imbalance via weighted sampling strategies during contextual example selection (Jang and Stikkel 2024). JNLP 2024 employs targeted few-shot prompting, dynamically optimizing prompt efficacy by evaluating multiple GLUE-derived templates on development sets, and selecting top-performing prompts to condition LLM outputs for enhanced entailment resolution in statute law applications (Nguyen et al. 2024d). FSW implements in-context learning using LLMs by comparing LLM-generated outputs against expert-annotated rationales through carefully designed demonstrations that explicitly incorporate legal reasoning chains and annotation rules, enabling the model to align its judgments with professional standards (Ma et al. 2024). AIIR 2025 dynamically adapts LLMs to legal textual entailment by leveraging few-shot demonstrations and zero-shot chain-of-thought prompting, enabling task specialization that enhances robustness through majority voting results across LLaMA and Mistral variants (Wu et al. 2025). NOWJ 2025 employs integrated in-context learning for legal case entailment by prompting LLMs with multiple candidate paragraphs under explicit judicial reasoning instructions, while dynamically retrieving similar queries via biencoder to address interpretation differences in case law applications (Nguyen et al. 2025a).

Small Models Legal retrieval scenarios impose distinctive constraints on SMs, where limited parameter capacity impedes the assimilation of intricate doctrinal hierarchies and contextually evolving precedent relationships. SMs frequently exhibit deficient recall in cross-jurisdictional retrieval and falter in reconciling semantically overlapping statutory provisions, particularly when handling rare cases or dynamically amended legislation. The integration of LLMs as auxiliary resources mitigates these constraints by serving as data sources that generate task-aligned synthetic datasets to alleviate data scarcity, and as supervision signals that guide SM training toward better domain adaptation. In the following, we analyze the use of LLMs as data sources and supervision signals for SMs.

As **data sources**, CAPTAIN 2024 employs LLMs to summarize legal premises and synthesize novel hypotheses, constructing augmented datasets that enable efficient fine-tuning of Flan-T5 via LoRA, thus enhancing statutory entailment resolution in low-resource settings (Nguyen et al. 2024b). LeDQA utilizes GPT-4 to generate sentence-level relevance annotations for downstream tasks, producing curated training corpora that refine the performance of TF-IDF and BERT-based retrieval models (Liu et al. 2024b). CaseLink utilizes the instruction-tuned e5-mistral-7b-instruct language model, adapted for legal retrieval tasks, to generate node embeddings from legal texts, constructing structured feature representations for cases and charges within the Global Case Graph, which serves as the foundational input for training downstream graph neural network modules optimized for case retrieval (Tang

et al. 2025). As **supervision signals**, LLMs distill latent knowledge to recalibrate SM learning objectives. CAPTAIN 2024 employs LLM-generated summarizations to synthesize novel training samples, subsequently utilizing these augmented data to fine-tune compact models via LoRA for COLIEE Task 4 (Nguyen et al. 2024b). JNLP 2024 integrates LLMs to refine top-5 candidate predictions from smaller models, by comparing LLM-generated results to guide the training of smaller models using existing data with the purpose of elevating their precision (Nguyen et al. 2024d). IRNLPUI 2025 employs relevance assessments generated by LLMs as discriminative features and combines them with BM25 scores and cosine similarities to optimize legal article selection for Japanese Civil Code retrieval (Tjandra et al. 2025).

Frameworks Traditional legal retrieval workflows face significant limitations in handling the field's inherent complexities, where rigid, linear processes fail to manage intricate legal relationships and evolving case law networks. Conventional systems typically rely on single-step matching over latent semantic relationships, ignoring iterative refinement and counterargument analysis crucial for comprehensive precedent discovery. Their "black box" nature further obscures retrieval pathways, preventing users from verifying relevance decisions, understanding how conflicting legal concepts are weighed, or identifying potential biases related to legal authority, recency, or interpretation. The emergence of LLMs transforms this landscape by reshaping retrieval into flexible, multi-stage processes that adapt dynamically to multi-jurisdictional contexts. These frameworks decompose complex retrieval tasks into context-responsive subtasks, enable iterative refinement through natural language interactions, and embed transparency by generating step-by-step reasoning chains that reference key legal concepts, factual similarities, and applicable norms. In the following analysis, we examine two roles: task workflow reshaping and explainability enhancement.

In the role of **task workflow reshaping**, CAPTAIN 2024 implements coarse-to-fine multi-stage re-ranking through LLMs, sequentially applying sparse retrieval, entailment scoring, and LLM-based distillation to optimize hierarchical relevance judgments across case law corpora (Nguyen et al. 2024b). GEAR integrates judgment prediction with retrieval operations using ChatLaw to generate law-aware hierarchical semantic IDs through traversing a law structure constraint tree, enabling dual predictions for both judgments and relevant documents in a single inference (Qin et al. 2024). RSUEQ extends "query-to-model" pathways into multi-phase reasoning sequences by leveraging LLMs for legal term inference and query reformulation, followed by tri-model ensemble voting to enhance cross-jurisdictional recall (Nguyen et al. 2024c). LawLLM, based on instruction-tuned Gemma-7B, reconfigures precedent recommendation as entity-linking tasks, transforming unstructured queries into knowledge-graph-guided multiple-choice formulations that align case facts with statutory entities (Shu et al. 2024). KELLER introduces a novel module called Knowledge-Guided Case Reformulation, decomposing raw case analysis into two-step crime-law extraction and structured sub-fact generation, enabling hierarchical semantic matching for interpretable relevance scoring (Deng et al. 2024b). JNLP 2024 transforms the traditional retrieval-to-entailment pipeline by strategically integrating LLMs at multiple stages, including re-ranking initial retrievals, refining high-recall predictions from smaller models, and directly adjudicating textual entailment via ensemble prompting, together contributing to the decomposition of complex tasks and enhancing end-to-end processing efficacy (Nguyen et al. 2024d). OVGU2 2025 employs LLM-generated legal propositions to reformulate case fragments into structured queries, enabling three-phase workflow decomposition: initial

BM25Plus retrieval, fragment-to-proposition transformation via phi4 and qwen2.5, and reranking using historical citation matrices for final candidate optimization (Wehnert et al. 2025).

In the role of **explainability enhancement**, LawLLM improves explainability in Precedent Case Recommendation tasks by prompting the model to output not only precedents but also the primary factors justifying their relevance to the query (Shu et al. 2024). KELLER improves explainability by demonstrating the quantitative contribution of each query and document sub-fact to the final relevance score, thus enhancing transparency and interpretability in the matching process by making sub-facts enumerable, alignable, and traceable to their sources of similarity (Deng et al. 2024b).

4.2.3 LLM for evaluation

Unlike general information retrieval, legal retrieval requires an evaluation framework that accounts for jurisdictional relevance, fidelity to precedents, and diversity of interpretation in the evolving context of case law. Traditional metrics, based on static relevance judgments and fixed semantic mappings, fail to capture these dimensions, limiting comprehensive assessment of retrieval results. LLMs address these challenges by modeling complex legal semantics and reasoning patterns, enabling more sophisticated evaluation paradigms such as structured scoring mechanisms and output reconstruction techniques. The following explores how LLMs advance evaluation by enhancing both dimensions and methods.

To extend **evaluation dimensions**, RSUEQ extends evaluation objectives by comparing whether terminology-enhanced queries or rewritten queries generated by LLMs lead to greater improvements in retrieval effectiveness, establishing new benchmarks for query formulation strategies (Nguyen et al. 2024c). DD-LLM pioneers evaluation methodologies by leveraging LLMs for pre-evaluation in few annotation or no annotation scenarios, thereby creating assessment frameworks for data-sparse environments (Jang and Stikkel 2024).

To enhance **evaluation methods**, LLMs enable sophisticated assessment paradigms in legal retrieval by applying output reconstruction techniques. Specifically, JNLP 2024 implements output reconstruction for evaluation by aggregating predictions from diverse LLM-prompt configurations through two ensemble techniques: majority voting, which selects the most frequent prediction across all configurations to consolidate outputs; and the Dawid-Skene model, which weighs inputs based on estimated reliability of each LLM-prompt combination to statistically infer latent ground truth, together transforming different model outputs into unified evaluation decisions (Nguyen et al. 2024d).

4.3 Legal generation

Legal generation is the task of automatically generating professional legal texts from diverse inputs, including legal articles, evidentiary records, case narratives, and analogous legal sources, according to specific legal requirements. It not only facilitates the generation of domain-specific legal artifacts, but also provides structured knowledge representations that serve as foundational components for advanced reasoning systems and decision-support frameworks (Thanh and Satoh 2024; Westermann 2024).

From a technical perspective, legal generation is a conditional text generation task where the input consists of various legal elements such as factual claims, legal articles,

or procedural terms, and the output is a coherent legal text, as illustrated in Fig. 8. In this architecture, a set of inputs from heterogeneous sources $E = \{e^1, e^2, \ldots, e^t\}$ first undergoes a *multi-source encoding and fusion step*, in which the heterogeneous inputs are jointly transformed into a *unified embedding representation* V. Unlike traditional encoder-decoder structures, this encoding process emphasizes *integration across diverse inputs*, and may adopt modular or hybrid designs, often leveraging techniques such as retrieval-augmented generation or domain-adaptive fine-tuning. The fused embedding V is then passed into the generator, which produces a hidden state sequence $Z = \{z^1, z^2, \ldots, z^t\}$. At each time step, the generator computes a probability distribution p over a predefined candidate set W, and the most probable token is selected to form the next output. This process is repeated autoregressively until the full legal text is generated.

Legal generation tasks mainly fall into the following four categories. **Legal judgment generation** takes case fact descriptions, applicable legal articles, and precedent materials as inputs, and outputs structured judicial documents including court views, rationales, and sentencing justifications, aiming to assist judges in drafting legally sound judgments (Epps et al. 2023; Deroy et al. 2024; Liu et al. 2024a; Xu et al. 2024; He et al. 2024). Legal question generation encompasses both query rewriting and question creating, transforming user queries with contextual and legal information into precise, coherent, and purpose-aligned questions. Query rewriting refines vague inputs, while question creating generates new questions to bridge knowledge gaps, thereby enhancing the accuracy and efficiency of legal information retrieval and interactive dialogues (Fung et al. 2024; Yuan et al. 2024). **Legal question answering** takes user queries with contextual details and legal knowledge bases as inputs, and outputs actionable legal advice or explanatory answers in interactive dialogue or long-form formats, designed to provide legal professionals and the public with efficient access to reliable legal information (Wiratunga et al. 2024; Louis et al. 2024; Bhatnagar and Huchhanavar 2025; Hu et al. 2025; Wahidur et al. 2025). **Legal document summarization** takes full legal documents as inputs and outputs concise, coherent summaries that preserve key information and legal reasoning, improving the accessibility and understanding of complex legal texts (Song et al. 2025; Mao et al. 2024).

LLMs are involved throughout the entire process of legal text generation tasks, driving innovations in model architecture and substantial improvements in generation quality. In the following, we provide a structured analysis of how LLMs contribute to LegalAI research across legal generation tasks. As shown in Fig. 9, we organize the roles of LLMs in legal generation tasks according to the taxonomy introduced in Table 1.

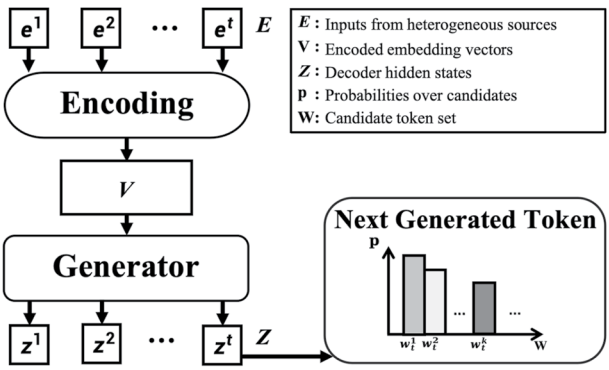


Fig. 8 The pipeline of legal generation tasks

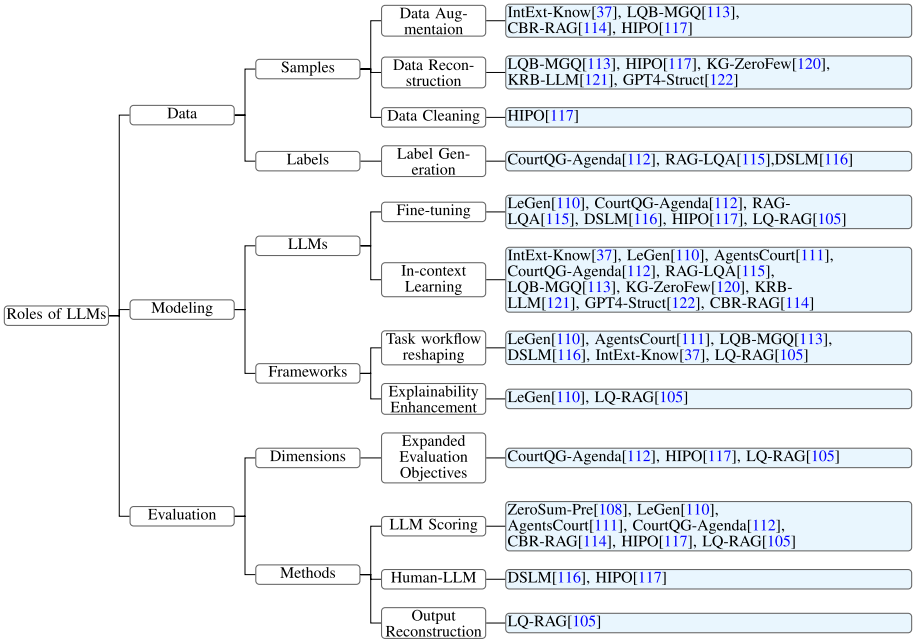


Fig. 9 Roles of LLMs in legal generation tasks

4.3.1 Data

We analyze how LLMs address data challenges across both sample and label objects.

Sample Legal generation at the sample level requires training data that is contextually grounded, stylistically aligned, and structurally faithful to the intended outputs. However, real-world legal corpora often suffer from domain sparsity, structural inconsistency, insufficient coverage of target generation formats, and a lack of paired input–output structures for supervised training. These shortcomings constrain models' ability to learn fine-grained semantics and adapt to diverse drafting scenarios. LLMs pretrained on legal corpora with strong generative abilities offer a promising solution. They can generate aligned input–output pairs that link case facts to rulings and legal articles to summaries, simulate diverse legal formats and reasoning styles, and reconstruct unstructured texts as structured representations such as event frames or logic-based rules. Furthermore, LLMs can clean noisy data by detecting hallucinated articles, incorrect references, and low-quality samples, and replace them with authentic, semantically consistent legal content. Collectively, these capabilities enrich stylistic and structural diversity, mitigate the scarcity of aligned samples, and ensure factual and semantic integrity, thereby enhancing the adaptability, robustness, and reliability of generative LegalAI systems in real-world applications. In the following, we analyze how LLMs contribute to current research through data augmentation, reconstruction, and cleaning.

In the role of **data augmentation**, IntExt-Know synthesizes charge clarification texts for legally ambiguous crime pairs such as drug trafficking versus drug possession within the charge clarification module. Through tailored prompts, LLMs generate comparative

analyses of key distinctions including criminal intent, quantity thresholds, and sentencing outcomes, which then undergo rigorous quality validation before being incorporated into an external knowledge repository, with the generated data further verified through human evaluation (Liu et al. 2024a). LQB-MGQ leverages GPT-3 to automatically generate 59,798 legal questions from CLIC texts through partitioned prompting. This LLM-driven augmentation expands question banks with large-scale and diverse queries, surpassing human capacity in both volume and cost efficiency, with the generated data subsequently evaluated across four dimensions, namely quantity, precision, coverage, and diversity, by human experts (Yuan et al. 2024). CBR-RAG utilizes GPT-4 to synthesize structured question-answer pairs from raw legal text snippets within the Australian Open Legal Corpus. By generating 2,124 novel QA triplets comprising a question, a supporting text segment and an answer, GPT-4 addresses data scarcity while preserving semantic alignment with source materials. The generated dataset is validated through manual review and similarity analysis to ensure factual accuracy, diversity, and coherence (Wiratunga et al. 2024). HIPO utilizes LLMs to generate novel legal questions and answers from authentic statutes, expanding the training dataset with 3,883 diverse QA pairs. The model-generated samples increase data variety while preserving semantic consistency and introducing linguistic diversity. These data undergo an automatic statute-correction procedure followed by expert verification, achieving over 80 % correction success and fewer than 4 % low-quality samples across separate evaluations (Hu et al. 2025).

In the role of **data reconstruction**, LQB-MGQ employs GPT-3 to reconstruct semantic representations by reformulating legal content into question formats. Through hybrid partitioning, it directs the LLM's attention per paragraph within sections, enabling perspective shifts like tenant versus landlord views without altering source material, thus restructuring knowledge presentation implicitly (Yuan et al. 2024). HIPO utilizes LLMs to refine generated legal question–answer pairs based on authentic legal articles, producing factually grounded and semantically coherent responses that constitute the fine-tuning dataset (Hu et al. 2025). KG-ZeroFew leverages GPT−3.5 to semantically reconstruct atomic sentences from the Spanish Workers' Statute into structured event triples. Through five-shot learning with optimized demonstrations, the LLM extracts Hohfeldian relations and argument roles, subsequently transforming them into an RDF knowledge graph using Provision Model and SEM ontologies (Argüelles Terrón et al. 2023). KRB-LLM processes legal text data using GPT-4o. Sequential prompts extract semantic elements from this input data, reconstructing them as executable Prolog rules. The conversion formalizes raw legal text data into symbolic predicates while prohibiting new fact creation, strictly binding outputs to predefined knowledge bases (Billi et al. 2024). GPT4-Struct leverages GPT-4 to reconstruct semantic data by processing legislative text to extract legal criteria and conclusions. It converts unstructured legal articles into structured JSON pathways that formally encode logical relationships between legal requirements and outcomes, enabling computable rule-based frameworks for expert systems (Janatian et al. 2023).

In the role of **data cleaning**, HIPO employs LLMs to clean generated legal responses through a multi-stage verification pipeline. First, it detects hallucinations including fabricated statutes, incorrect code numbers, or irrelevant legal articles. For identified errors, the system retrieves semantically aligned authentic articles from a structured legal database using embedding-based similarity matching. These flawed elements are then replaced with

validated references. Simultaneously, low-quality data such as overly brief questions under 20 words or logically inconsistent answers undergo automated filtration (Hu et al. 2025).

Label Legal generation tasks at the label level depend on precise, semantically aligned annotations that guide models to produce outputs in the correct format and style. Yet real-world legal corpora rarely contain such fine-grained or clearly defined labels, whether for output type, rhetorical style, intended audience, or explanatory intent, which limits supervised training and leads to inconsistent outputs. LLMs offer a scalable solution by automatically generating structured labels, resolving ambiguities, and enriching incomplete annotations, thereby improving alignment and consistency. However, their use in label-level generation remains limited, as research in this area has largely emphasized improving the quality and validity of generated content, while the absence of standardized label taxonomies makes label-based supervision less practical and harder to evaluate. Several recent works demonstrate the role of **label generation** in practice. CourtQG-Agenda employs GPT-3 as a zero-shot oracle to produce silver-standard natural language labels for agenda explanations. This process inputs courtroom dialogue context and witness responses into GPT-3 prompts, generating intent-revealing labels such as exposing witness bias or highlighting testimony inconsistencies. These machine-generated annotations are described as silver-standard labels that support the subsequent training of subsequent models (Fung et al. 2024). RAG-LQA uses LLMs to synthesize rationales at the paragraph level by processing all paragraphs from relevant legal articles per question. It identifies evidentiary paragraphs contributing to answers, outputting comma-separated identifiers or "None". Generated rationales are directly incorporated into LLeQA without refinement, enabling scalable interpretable supervision for downstream tasks (Louis et al. 2024). DSLM leverages GPT−3.5 to generate descriptive labels for 119 legal intents identified by BERTopic. It processes topic documents and keywords to generate concise, human-readable intent names such as "Landlord Lease Deposit Disputes" through a structured prompt (Bhatnagar and Huchhanavar 2025).

4.3.2 Modeling

We analyze how LLMs contribute to legal generation modeling across three objects, namely LLMs, small models and frameworks.

LLMs Compared to general text generation tasks, legal generation presents unique challenges due to its demand for stylistic precision, structural fidelity, and domain-specific reasoning. Legal documents often exhibit rigid formatting, jurisdictional variation, and specialized rhetorical structures that are difficult for general-purpose LLMs to replicate. Furthermore, legal generation encompasses a wide range of task types, including question answering, rationale synthesis, judgment drafting, and agenda explanation, each requiring distinct reasoning patterns and strict adherence to judicial writing styles and factual accuracy. Traditional full-parameter fine-tuning is computationally expensive, prone to overfitting on narrow domains, and may degrade general language understanding, while fixed-parameter models struggle with long contexts, logical consistency, and accurate citation practices. LLMs help address these issues by enabling parameter-efficient fine-tuning and multi-stage pipelines that specialize models for legal generation without excessive resource costs, as well as advanced prompting strategies that tailor generation to specific case scenarios. Through few-shot prompting, chain of thought reasoning, retrieval aug-

mented generation, and prompt refinement, LLMs can ground outputs in authoritative legal sources, reduce hallucination, and maintain coherence across complex legal structures. These methods collectively strengthen alignment with legal semantics, factual precision, and task-specific structures, establishing LLMs as a promising foundation for robust and adaptable legal generation systems. In the following, we specifically analyze fine-tuning and in-context learning.

In the role of **fine-tuning**, within the LeGen framework, the LeGen-FT variant employs parameter-efficient fine-tuning via Low-Rank Adaptation to specialize the LLM for legal text generation. This adaptation optimizes weights using curated instruction datasets that map legal concepts to rationales, adjusting specific layers (rank=16, alpha=64) to internalize juridical patterns while preserving linguistic generality (Xu et al. 2024). CourtQG-Agenda fine-tunes LLMs on legal corpora for three specialized tasks: agenda explanation generation, question type prediction, and dialogue-aware question generation. The models undergo domain adaptation through supervised training on courtroom transcripts and complaint documents, modifying their parameters to internalize legal reasoning patterns (Fung et al. 2024). RAG-LQA applies QLoRA-based fine-tuning to LLaMA on the LLeQA dataset. The base weights are quantized in 4-bit NormalFloat format, while all original model parameters are frozen. Trainable 16-dimensional LoRA adapters, with scaling factor 32, inject task-specific parameters into all linear layers of the frozen model. Optimization uses 4-bit paged AdamW on LLeQA's question-provision-answer triples, minimizing the negative log-likelihood for statutory response generation (Louis et al. 2024). DSLM framework drives the Mistral 7B model to perform domain-adaptive fine-tuning through two critical phases. First, supervised fine-tuning on the "legal-qa-dataset" to internalize legal response patterns. Second, direct preference optimization using human-ranked "legal-qa-preference-dataset" outputs aligns responses with expert-level precision (Bhatnagar and Huchhanavar 2025). HIPO implements a specialized two-phase fine-tuning strategy to improve legal factuality of LLMs. Initially, supervised fine-tuning trains the model on automatically constructed legal QA datasets, minimizing negative log-likelihood loss to teach accurate statute citation. Second, Hard Sample-aware Iterative Direct Preference Optimization refines responses by optimizing preferences for hallucination mitigation. This enhances the model's ability to generate factual answers in legal question answering (Hu et al. 2025). LQ-RAG fine-tunes LLMs using domain-specific legal QA datasets and general instruction datasets. It employs Low-Rank Adaptation to optimize the generative LLM, reducing trainable parameters while enhancing domain adaptation and instruction-following capabilities. The fine-tuned models are merged into a Hybrid Fine-tuned LLM for specialized response generation (Wahidur et al. 2025).

In the role of **In-context learning**, IntExt-Know retrieves similar past cases according to the "treat-like-cases-alike" principle and supplies them as in-context examples for the LLM (Liu et al. 2024a). Within the LeGen framework, the LeGen-PT operates as zero-parameter legal reasoners via pure in-context learning. The Solver module utilizes LLMs to generate concept-specific rationales from fact descriptions that are augmented with similar case demonstrations. Subsequently, the Generator module synthesizes court views by integrating verified rationales and case facts through structured prompting. This approach leverages LLMs' inherent juridical knowledge retrieval capabilities exclusively through API-based contextual conditioning, transforming generic models into legal assistants without weight modification (Xu et al. 2024). AgentsCourt leverages the in-context learning capabilities of

GPT models to simulate legal roles without fine-tuning. The judge, plaintiff, and defendant agents are dynamically contextualized through role-specific prompts derived from real case documents. Additionally, the judge assistant retrieves and synthesizes legal resources using task-oriented prompts, adapting knowledge application to case contexts (He et al. 2024). CourtQG-Agenda integrates semantically relevant sentences from complaint files and witness descriptions into zero-shot prompts for LLMs. Models such as GPT-3 dynamically generate court examination questions conditioned solely on these background-derived textual cues (Fung et al. 2024). RAG-LQA applies in-context learning using instruction-tuned LLaMA-derived LLMs. These models process structured inputs comprising task descriptions, top-5 retrieved Belgian legal articles, and dynamically selected demonstrations. The zero-shot, one-shot or few-shot configurations incorporate question-answer examples retrieved by CamemBERT. Responses are generated automatically without parameter updates, using 4-bit quantization for extended context windows (Louis et al. 2024). LQB-MGQ leverages GPT-3's in-context learning capability by feeding partitioned CLIC text segments with instructional prompts. Without fine-tuning, the LLM dynamically interprets legal contexts to generate relevant queries, demonstrating zero-shot adaptation to specialized domains through implicit task recognition (Yuan et al. 2024). KG-ZeroFew employs GPT−3.5 using in-context learning for event extraction. It dynamically adapts to the task using carefully engineered prompts containing task-specific demonstrations. The method employs zero-shot, 3-shot, and optimized 5-shot learning strategies, leveraging the model's ability to infer patterns directly from examples within the prompt context (Argüelles Terrón et al. 2023). KRB-LLM employs in-context learning through a dynamic "Chain of Prompts" methodology, utilizing GPT-4o with few-shot demonstrations from legal articles and case law. For article formalization, it iteratively refines Prolog rules via three-step prompting. For case law expansion, it formalizes case texts into Prolog rules via sequential prompts (Billi et al. 2024). GPT4-Struct leverages in-context learning by processing legislative text through task-specific system messages without fine-tuning. Iteratively optimized prompts guide the model to parse legal criteria and conclusions into structured pathways. Operating in a zero-shot manner, it reconstructs complex legal semantics solely from contextual instructions (Janatian et al. 2023). CBR-RAG leverages Mistral-7B for in-context learning by dynamically retrieving relevant legal cases and injecting them as contextual prompts. This augmented input enables the LLM to generate factually grounded answers without fine-tuning, directly utilizing case-based reasoning to enhance domain-specific knowledge retrieval and response accuracy in legal QA tasks (Wiratunga et al. 2024).

Small Models In contrast to legal classification and retrieval tasks where LLMs are frequently used to augment SMs by generating extra training data or distilling knowledge to compensate for limited data and model capacity, legal text generation has seen little research on large-small model collaboration. Modern LLMs can already perform complex generative tasks like drafting judgments or summarizing cases independently, which diminishes the need for a SM's assistance. Empirical evidence shows that LLM-driven approaches yield substantial performance gains in legal generation tasks even without an auxiliary SM, suggesting that an LLM alone is often sufficient (Deroy et al. 2024; Katz et al. 2024; Benedetto et al. 2025). Moreover, technical factors make multi-model generation less appealing. Unlike classification with discrete labels or retrieval producing document rankings, where LLM outputs can directly aid in training SMs, text generation requires sequential decoding with strong token interdependency. Splitting or coordinating the generation process between

LLMs and SMs complicates decoding and risks disrupting coherence because any hand-off in the middle of a sentence can propagate errors or inconsistencies. Handling prompts and intermediate outputs from two models also adds overhead and potential error accumulation. Given that a single LLM can generate fluent, contextually accurate legal text and reasoning, researchers have had less incentive to pursue collaborative frameworks for generation tasks. Indeed, the specifics of such LLM–SM cooperative decoding remain largely unexplored in the broader NLP literature and this gap is especially pronounced in the legal domain (Zhang et al. 2024).

Frameworks Compared to general generation tasks, legal generation encompasses intricate workflows with multi-step reasoning, domain logic, and context-sensitive output structures. Traditional one-shot approaches often treat the task as a static input–output mapping, failing to capture sequential complexity, adapt to evolving case specifics, or reveal the reasoning process underlying outputs. LLM-based frameworks overcome these limitations by decomposing tasks into coherent subtasks, chaining reasoning steps, and embedding natural language rationales within the workflow. This modular and flexible orchestration enables subtasks such as fact identification, rationale synthesis, and drafting to be dynamically coordinated, while also exposing intermediate reasoning for greater transparency, error tracing, and legal soundness. In the following, we specifically analyze two roles: task workflow reshaping and explainability enhancement.

In the role of **task workflow reshaping**, LeGen serves as the core engine within a novel divide-and-conquer paradigm for restructuring the court view generation process. The framework decomposes monolithic generation into four specialized modules. First, the Decomposer identifies key legal concepts. Then, LLMs in the Solver independently generate concept-anchored rationales using contextual prompts enriched by similar cases. Subsequently, LLMs in the Generator synthesize these verified rationales into the final court view. This modular orchestration explicitly leverages the contextual reasoning capabilities of LLMs, replacing end-to-end generation with a structured, concept-driven workflow (Xu et al. 2024). AgentsCourt reshapes judicial workflow into an LLM-driven pipeline by decomposing traditional court procedures into three specialized agent modules. It transforms the sequential stages of opening, debate, retrieval, and judgment into parallelized multi-agent collaboration through debate simulation, legal resource retrieval, and judgment refinement modules. Each module delegates distinct tasks to LLM agents, which optimizes judicial efficiency while rigorously preserving procedural integrity (He et al. 2024). LQB-MGQ employs GPT-3 to reshape the LQB construction workflow by dynamically partitioning CLIC-pages into sections and paragraphs, generating context-rich prompts for machine-question generation. It transforms manual curation into an adaptive pipeline for legal knowledge dissemination (Yuan et al. 2024). DSLM reshapes the legal assistance pipeline by integrating LLMs into three transformative phases. First, BERTopic-GPT3.5 automates intent classification from raw Reddit data. Second, Mistral 7B undergoes supervised fine-tuning (SFT) and direct preference optimization (DPO) to become a domain-specific legal expert. Finally, the DSLM-augmented GPT3.5 generates responses guided by mined intents. This end-to-end framework establishes a viable alternative to retrieval-augmented generation techniques by implementing contextual domain adaptation (Bhatnagar and Huchhanavar 2025). IntExt-Know decomposes the monolithic generation process into three specialized submodules by leveraging the contextual reasoning of GPT models. The models generate comparative charge clarifications to stimulate internal legal knowledge, interpret

retrieved statutory provisions for precise article citation, and extract sentencing elements to invoke executable Python functions for symbolic prison-term computation (Liu et al. 2024a). LQ-RAG reshapes task workflows through a recursive feedback mechanism. The GPT-4 evaluation agent assesses responses against three metrics, triggering a prompt engineering agent to dynamically reformulate queries when thresholds are unmet. This iterative loop replaces linear RAG execution with adaptive cycles of retrieval-regeneration-evaluation, fundamentally restructuring legal QA processes (Wahidur et al. 2025).

In the role of **explainability enhancement**, LeGen prompts LLMs to generate concept-anchored rationales within this divide-and-conquer workflow. By decomposing the court view generation into concept-specific sub-views, the framework makes each reasoning step explicit and traceable. The Solver articulates legal rationales for individual concepts, and the Generator integrates them into a coherent final decision, thereby revealing how each legal concept contributes to the judgment and improving the transparency and interpretability of the overall process (Xu et al. 2024). LQ-RAG enhances explainability within its generative workflow through a structured Chain-of-Thought–based evaluation mechanism. The GPT-4–powered evaluation agent decomposes each generated response into discrete claims, cross-verifies them against retrieved legal contexts, and assesses their relevance and factual grounding. This structured, step-wise reasoning process makes the evaluation of legal QA outputs transparent and traceable, thereby improving the interpretability of generation quality decisions (Wahidur et al. 2025).

4.3.3 Evaluation

Compared with general text generation, legal generation presents distinct evaluation challenges arising from the open-ended nature of outputs, the demand for factual precision, adherence to stylistic conventions, and alignment with normative standards. Traditional evaluation methods, whether based on fixed references (e.g., BLEU, ROUGE) or surface-level metrics (e.g., perplexity, n-gram overlap), often fail to capture these complexities, providing little insight into content fidelity, factual grounding, or legal validity. They also overlook critical dimensions such as faithful citation, structured argumentation, and multi-turn reasoning, while human evaluation protocols remain inconsistent, fragmented, and resource-intensive. LLMs provide a path forward by enabling richer and more adaptive evaluation frameworks. Acting as automated judges, they can assess fluency, factual accuracy, logical coherence, and adherence to legal standards through structured prompts and context-sensitive reasoning. Moreover, when combined with human oversight, LLMs can standardize initial assessments, highlight areas of concern, and support hybrid protocols that blend quantitative and qualitative insights. Together, these approaches extend evaluation beyond correctness to encompass semantics, structure, and normative soundness, offering more reliable and scalable assessments of legal generation models. The following explores how LLMs advance evaluation by enhancing both dimensions and methods.

To extend **evaluation dimensions**, CourtQG-Agenda introduces agenda-awareness assessment through GPT-4–driven scoring by automatically judging whether the generated courtroom questions align with either direct or cross-examination stances (Fung et al. 2024). HIPO employs LLMs to establish multi-dimensional factuality assessment in legal question answering. The LLM directly generates and scores novel evaluation metrics, including Non-Hallucinated Statute Rate, Statute Relevance, and Legal Claim Truthfulness, enabling

automatic evaluation of unlabeled outputs and extending assessment targets beyond surface-level accuracy (Hu et al. 2025). LQ-RAG integrates GPT-4 as an evaluation agent to assess answer relevance, context relevance, and groundedness. Through a Chain-of-Thought–based reasoning process, the agent decomposes responses into discrete claims and searches the retrieved context for supporting evidence. This tripartite evaluation framework moves beyond conventional single-metric assessment, establishing domain-specific rigor in evaluating legal QA outputs (Wahidur et al. 2025).

To enhance **evaluation methods**, LLMs enable sophisticated assessment paradigms in legal generation by facilitating structured scoring mechanisms, Human–LLM collaboration, and output reconstruction techniques. Specifically, in the role of **scoring**, ZeroSum-Pre serves to compute perplexity scores during the evaluation. This automatic metric measures the fluency of generated summaries, employing both the GPT-2 model with its sliding window technique and the summary-generating models themselves (Epps et al. 2023). LeGen incorporates GPT-4 as an automated evaluator for generated court views through quantitative scoring on a scale from 1 to 5. Using a standardized prompt template, GPT-4 assesses output quality by holistically comparing model-generated texts with ground-truth references, emphasizing legal professionalism and rationale coherence alongside fluency (Xu et al. 2024). AgentsCourt employs GPT-4 as automated evaluators to quantitatively assess judicial outputs, which analyze civil and administrative judgments by decomposing complex verdicts into discrete key points comparing them against reference standards. It calculates precision and recall metrics based on semantic alignment of legal elements such as compensation amounts, penalty terms and dismissal clauses (He et al. 2024). CourtQG-Agenda deploys GPT-4 to automatically compute Agenda Scores by assessing whether generated questions reflect direct or cross-examination stances, which analyzes question texts and outputs tokens as quantitative measures of agenda alignment (Fung et al. 2024). CBR-RAG utilizes Mistral-7B to generate embeddings for its own answers and reference responses. Cosine similarity quantifies semantic alignment between these embeddings, automating quality assessment without human intervention (Wiratunga et al. 2024). HIPO utilizes LLMs as automated evaluators to score legal responses using three hallucination metrics: Non-Hallucinated Statute Rate, Statute Relevance Rate and Legal Claim Truthfulness. GPT-4 turbo computes these scores by aligning generated content with reference statutes and human-annotated benchmarks, validating the reliability of the metric via a high Spearman correlation with human judgments (Hu et al. 2025). LQ-RAG deploys its GPT-4-powered evaluation agent to rigorously score responses. It verifies context relevance by cross-checking retrieved passages, assesses groundedness through claim decomposition and evidence alignment, and measures answer relevance against the original query. This tripartite validation ensures factual integrity (Wahidur et al. 2025). In the role of **Human-LLM**, DSLM deploys GPT-4 as the primary automated evaluator to score responses from Mistral 7B variants and GPT−3.5 using chain-of-thought prompting. The LLM assesses relevance, accuracy, and detail depth on a scale from 1 to 10 across 100 test samples, calculating win rates against human references. Human experts then validate critical legal nuances in 46 responses, identifying GPT-4's limitations in discerning jurisprudential subtleties. This hybrid approach quantifies model superiority while exposing evaluation gaps (Bhatnagar and Huchhanavar 2025). HIPO establishes a human-AI collaborative framework where legal experts curate LegalHalBench's QA data and annotate training samples, while LLMs generate initial responses. Lawyers subsequently verify statutory accuracy or relevance to

resolve discrepancies in hallucination labels. This synergy ensures the credibility of the data set, as human oversight corrects LLM errors and finalizes evaluation standards (Hu et al. 2025). In the role of **output reconstruction**, LQ-RAG employs a GPT-4–based feedback loop for evaluation. When the evaluation agent detects insufficient relevance or groundedness, it initiates regeneration via a prompt engineering agent, ensuring that the final output logically aligns with contextual evidence and legal reasoning, thereby enhancing the rigor and completeness of evaluation (Wahidur et al. 2025).

4.4 Emerging legal scenarios

The development of LLMs has given rise to several new legal task scenarios. In **overruling prediction**, the difficulty of modeling implicit legal relationships between cases and capturing discourse-level dependencies is addressed through LLMs' ability to process long contexts, integrate evidence from multiple legal documents, and derive structured representations of legal relations. Given the full texts or summaries of a citing case and a cited case, the model outputs a binary label indicating whether the citing case overrules the cited one (Sheik et al. 2024). In **legal knowledge generation**, LLMs are employed to extract, synthesize, and structure legal norms or reasoning patterns from unstructured legal texts, addressing the challenge of implicit rule identification and knowledge abstraction across diverse legal domains. Leveraging LLMs' ability to process lengthy documents and generalize over heterogeneous inputs, the model takes a raw legal article, case opinion, or statutory paragraph as input and generates an interpretable representation of its normative content for symbolic AI systems (Argüelles Terrón et al. 2023; Billi et al. 2024; Janatian et al. 2023). In **courtroom simulation**, LLMs act as legal agents and engage in deliberative reasoning by comparing competing contextual signals, reconciling conflicting interpretations, and generating step-by-step explanatory justifications before reaching a decision (Jung and Jung 2025; Chen et al. 2025).

5 Quantitative analysis across LLM roles and legal tasks

To evaluate the practical performance of LLMs in LegalAI, this section examines their effectiveness across legal tasks, including legal classification, generation, and retrieval. We focus on modeling-related roles, as they directly capture how LLMs contribute to method design and implementation, and these contributions largely accounts for the observed performance differences compared to non-LLM approaches. Meanwhile, the role of explainability enhancement, when it does not affect performance, is therefore not considered. We selected representative studies with clearly described experimental settings and explicit baseline comparisons to ensure fair and consistent comparison across studies by measuring the performance differences relative to each paper's own baseline. Specifically, we analyze the impact of LLMs in different roles within the same task and their performance on different tasks in the same role.

5.1 Comparison between different roles

In this section, we analyze performance improvements achieved by LLMs across different roles within the same task.

Legal Classification From Table 2, we have the following observations for legal classification tasks[3]: (1) Among different modeling roles, the combination of framework reshaping and LLM enhancement yields the most consistent and stable improvements across classification tasks. This indicates that integrating LLMs into task-level modeling structures more effectively leverages their capabilities. In contrast, LLM enhancement alone leads to more limited and less consistent gains. (2) Legal charge prediction shows substantial and consistent gains across most metrics, indicating that its clear label structure enables LLMs to better align inputs with legal outcomes. Legal article prediction shows moderate benefits, with several methods achieving improvements, although performance drops are observed (e.g., PLJP), likely due to multi-label complexity and the reasoning demands of legal articles. Prison term prediction shows the least consistent improvement, reflecting the ambiguity of sentencing boundaries and the weaker correlation between input descriptions and target outputs. (3) Currently, with the involvement of LLMs, applicability classification has been

Table 2 Performance comparison of methods on legal classification tasks, with color intensity indicating the magnitude of change: red (+) for performance improvement and blue (-) for performance degradation

Task	Method	Datasets	Roles	Metrics			
				Accuracy	MaF	MaP	MaR
LAP	MPEG[73]	FA2019 T2	LE	–	38.71 (+5.44)	42.43 (+8.4)	53.78 (+10.09)
	PLJP[34]	CAIL2018	FR,LE	87.07 (-0.75)	56.63 (+0.82)	58.81 (+2.68)	57.29 (+0.72)
	LegalReasoner[69]	CAIL2018	FR,LE	89.30 (+2.20)	88.90 (+2.20)	–	–
	LegalReasoner[69]	ECtHR	FR,LE	85.50 (+2.80)	85.10 (+3.00)	–	–
	ADAPT[65]	CAIL2018	FR,LE	78.30 (+2.70)	80.00 (+2.30)	–	–
	DPFSI[70]	CAIL2018	LE	87.83 (+2.88)	85.33 (+2.4)	86.72 (+2.81)	85.48 (+2.16)
LCP	PLJP[34]	CAIL2018	FR,LE	94.99 (+3.45)	91.33 (-0.25)	92.12 (+0.51)	91.10 (-0.86)
	ADAPT[65]	CAIL2018	FR,LE	77.90 (+3.10)	83.00 (+3.70)	–	–
	DPFSI[70]	CAIL2018	LE	90.23 (+0.58)	89.71 (+0.75)	88.85 (+0.11)	89.06 (+0.16)
	CADLRA[66]	CAIL_CAD	FR,LE	82.82 (+10.98)	–	–	–
PTP	PLJP[34]	CAIL2018	FR,LE	48.72 (+7.88)	35.43 (-3.07)	42.64 (+4.02)	36.80 (-1.75)
	ADAPT[65]	CAIL2018	FR,LE	37.90 (+4.00)	35.80 (+2.30)	–	–
	DPFSI[70]	CAIL2018	LE	43.19 (+0.18)	40.75 (-0.29)	41.65 (-0.29)	39.76 (-0.3)
App.	GOLDCOIN[64]	HIPAA PR	FR,LE	99.53 (+5.14)	99.53 (+5.14)	–	–
Comp.	GOLDCOIN[64]	HIPAA PR	FR,LE	76.64 (+2.81)	64.83 (+2.43)	–	–

The roles of LLMs are categorized as "LLM Enhancement (LE)" and "Framework (FR)", representing roles under the objects "LLMs" and "Frameworks", while the object "Small Models" is not included as it is not involved in the papers selected for quantitative analysis. Legal classification tasks include Legal Article Prediction (LAP), Legal Charge Prediction (LCP), Prison Term Prediction (PTP), Applicability Classification (App.), and Compliance Classification (Comp.). The metrics used include Accuracy, Macro-F1 (MaF), Macro Precision (MaP), and Macro Recall (MaR). Values in parentheses represent the improvement or decline compared to the best results of the baseline models

[3] Note that in LJP, the Chinese dataset CAIL has been explored most extensively. This is primarily because a large proportion of LLM-based legal judgment prediction research has been conducted by Chinese researchers, and the CAIL dataset is explicitly divided into three subtasks: legal article prediction, legal charge prediction, and prison term prediction, making it widely adopted in LJP studies.

relatively well addressed. While legal article prediction and legal charge prediction perform well in most cases, their effectiveness still varies under different experimental settings, such as differences in language or legal systems. In contrast, prison term prediction and compliance classification still show significant room for improvement.

Legal Retrieval From Table 3, we have the following observations for legal retrieval tasks: (1) Framework reshaping is the most frequently adopted modeling role in the reviewed methods and generally leads to notable and stable improvements across retrieval tasks. In comparison, LLM enhancement and hybrid roles appear less effective. Performance drops are observed in certain tasks, such as litigation document retrieval, suggesting that its effectiveness may depend on task complexity and the required level of relevance. (2) The impact of LLMs varies across retrieval scenarios. Legal article retrieval benefits from LLMs, with improvements observed across all metrics, especially when query expansion is incorporated and external legal knowledge is introduced. Legal case retrieval also shows mostly positive gains, particularly in ranking-based metrics such as NDCG and top-k accuracy, reflecting the suitability of LLMs for deep semantic matching and context-aware relevance scoring. In contrast, the performance of litigation document retrieval varies, as recall improves substantially while precision suffers, indicating that LLMs still struggle to distinguish highly relevant documents from less related ones in this setting. (3) Overall, all retrieval tasks still exhibit considerable room for improvement. Among them, legal article retrieval and legal case retrieval show relatively better performance. Other tasks, including litigation document retrieval, similar case retrieval, and precedent recommendation, remain less mature and require further development.

Legal Generation From Table 4, we have the following observations for legal generation tasks: (1) LLM enhancement is more widely adopted across legal generation methods

Table 3 Performance comparison of methods on legal retrieval tasks, with color intensity indicating the magnitude of change: red (+) for performance improvement and blue (-) for performance degradation

Task	Method	Datasets	Roles	Metrics			
				P	R	MRR	Other metrics
LAR	RSUEQ[90]	COLIEE23	FR	74.78 (+2.51)	82.18 (+3.47)	–	–
	LQ-RAG[105]	Eva. data 1-6	LE	–	–	39.87 (+5.54)@5	Hit Rate@5: 51.45 (+6.04)
	KG-LLM[127]	CLAKG	FR	–	–	–	Acc: 69.40 (+40.50)
CLR	JNLP 2025[94]	COLIEE25	FR,LE	43.1(+5.24)	42.38(-1.35)	–	–
	KELLER[99]	LeCaRDv2	FR	63.13 (+8.55)@3	–	–	NDCG@3: 84.97 (+6.30)
	FSW[98]	LeCaRD	LE	–	–	–	NDCG@30: 94.58 (+4.39)
	LawLLM(SCR)[35]	CaseLaw	FR,LE	–	–	–	top3: 63.20 (+5.30)
	LawLLM(PCR)[35]	CaseLaw	FR,LE	–	–	–	top3: 59.70 (+8.30)
LDR	DD-LLM[103]	KIRA	LE	72 (-14.00)	96 (+25)	–	–
	GEAR[104]	ELAM	FR	–	23.68 (+2.36)@5	23.65 (+3.36)	–

The roles of LLMs are categorized as "LLM enhancement (LE)" and "Framework (FR)". Legal retrieval tasks include Legal Article Retrieval (LAR), Case Law Retrieval (CLR), and Litigation Document Retrieval (LDR). Metrics used include Precision (P), Recall (R), Mean Reciprocal Rank (MRR), Hit Rate, Accuracy (Acc.), normalized discounted cumulative gain (NDCG), and Top-3 Accuracy (top-3). The notation "@k" indicates the metric calculated over the Top K retrieved results (i.e., values like "23.68 (+2.36)@5" under Recall represent Recall calculated on the top 5 results). To avoid overcrowding, some metrics that apply only to specific methods are listed under the "Metrics" column

Table 4 Performance comparison of methods on legal generation tasks, with color intensity indicating the magnitude of change: red (+) for performance improvement and blue (-) for performance degradation

Task	Method	Datasets	Roles	Metrics				
				MaF/MiF	Precision	Rouge-L	BLEU-N	Other metrics
LJG	AgentsCourt[111]	SimuCourt	FR,LE	–	–	–	–	–
	IntExt-Know(LCG)[37]	LAIC2021	FR,LE	MiF: 50.70 (+3.60)	–	–	–	–
	IntExt-Know(LAG)[37]	LAIC2021	FR,LE	MaF: 93.12 (+2.16)	–	–	–	–
	AgentsCourt(LAG)[111]	SimuCourt	FR,LE	MiF: 93.73 (-2.61)	–	–	–	–
	IntExt-Know(PTG)[37]	LAIC2021	FR,LE	MaF: 65.18 (+5.08)	–	–	–	–
	IntExt-Know(CVG)[37]	LAIC2021	FR,LE	MiF: 83.31 (-9.84)	–	–	–	–
	LeGen-PT(CVG)[110]	LCVG	FR,LE	–	–	–	–	–
	AgentsCourt(LCAG)[111]	SimuCourt	FR,LE	MiF: 20.30 (+6.00)	–	–	–	Accuracy: 49.26 (+6.89)
				MaF: 44.12 (+7.60)				
LQA	RAG-LQA[115]	LLeQA	LE	–	–	–	–	METEOR: 19.70 (+3.50)
	CBR-RAG[114]	ALQA	LE	–	–	–	–	Cos scores: 91.41 (+1.74)
	DSLM[116]	legal-qa-dataset	FR,LE	–	–	–	–	WR(Max): 54.80 (+31.50)
	LQ-RAG[105]	Legal-QA et al.	FR,LE	–	–	–	–	Groundedness: 82.00 (+6.00)
	HIPO[117]	CAIL2018	LE	–	–	34.00(+7.40)	–	BERTScore: 76.20 (+2.30)
LQG	CourtQG-Agenda[112]	Bloomberg Law et al.	LE	–	–	23.40 (+3.40)	–	SimSc: 34.90 (+3.70)
	LQB-MGQ[113]	CLIC-pages	LE	–	68.00 (+18.00)	–	–	Coverage: 93.00 (+22.50)
				–	–	66.56 (+0.96)	57.42 (+1.65)	–
				MaF: 80.56 (+6.06)	79.63 (+3.80)	43.02 (+14.01)	52.67 (+1.77)	Recall: 80.67 (+7.31)
				–	–	–	–	Correctness: 55.00 (+2.70)

The roles of LLMs are categorized as "LLM enhancement (LE)" and "Framework (FR)". Legal generation tasks include legal judgment generation (LJG), which consists of charge generation (LCG), article generation (LAG), prison term prediction (PTG), court view generation (CVG), and case analysis generation (LCAG), along with legal question answering (LQA) and legal question generation (LQG). Metrics include Macro-F1 (MaF), Micro-F1 (MiF), Precision, Recall, ROUGE-L (based on Longest Common Subsequence), BLEU-N (n-gram precision–based metric), Accuracy, Correctness, METEOR, Cosine Similarity Score (Cos scores), Win Rate (Maximum) (WR(Max)), Groundedness, BERT-based Similarity Score (BERTScore), Similarity Score (SimSc), and Coverage

and demonstrates relatively stable and consistent improvements across generation tasks. In contrast, the combination of framework reshaping and LLM enhancement yields mixed outcomes, with some tasks showing notable metric gains while others exhibit performance drops. (2) The impact of LLMs varies across generation scenarios, but overall they consistently improve performance. Legal judgment generation benefits from LLMs, with notable gains in both accuracy and expressiveness. Legal question answering also achieves consistent improvements, as LLMs enhance reasoning quality and factual grounding. Legal question generation exhibits the most substantial advances, particularly in coverage and semantic consistency, underscoring LLMs' strength in reformulating inputs while preserving meaning. (3) Currently, legal charge generation shows the highest task completion, with very high baseline scores and only limited gains from LLMs. In contrast, court view generation benefits the most from LLM integration, with improvements across metrics. Legal question answering performs relatively well in several settings, while prison term generation, legal article generation, case analysis generation, and legal question generation remain less mature, showing lower absolute scores overall but clear improvements under LLM integration.

Overall, these results highlight that within each legal task type, the effectiveness of LLMs varies significantly across modeling role. Across classification and retrieval tasks, framework reshaping yields the most robust and consistent improvements, especially for tasks involving complex reasoning or multi-label structures. LLM enhancement brings greater

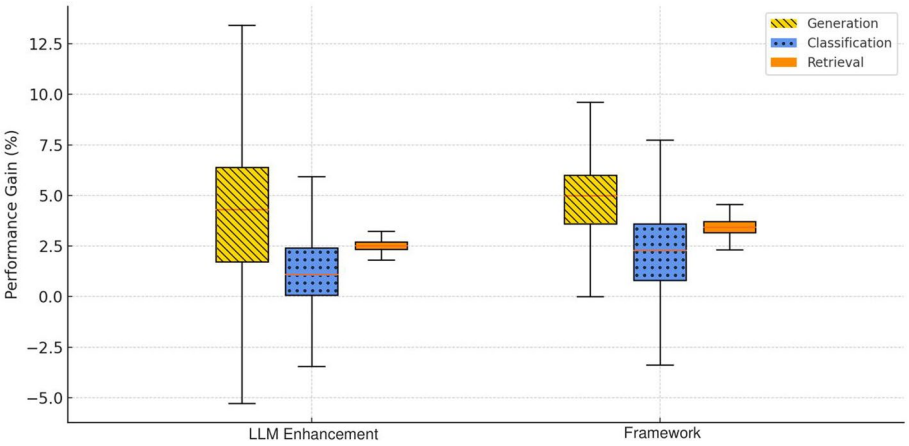


Fig. 10 Performance gains across LegalAI tasks under different modeling roles. Each box represents the distribution of relative performance gains (in F-score-based metrics) for a specific task type and modeling role. The horizontal orange line inside the box indicates the median, while the top and bottom edges correspond to the upper (Q3) and lower (Q1) quartiles, representing the interquartile range (IQR). The whiskers extend to the most extreme values within 1.5 × IQR from the quartiles, excluding potential outliers

and more stable performance gains in generation tasks, but it also remains valuable for classification and retrieval tasks with well-defined inputs and outputs, although its impact may be constrained by task complexity. Hybrid strategies, which combine both modeling roles, often achieve a balance between performance gains and stability and are widely applied across all three tasks, though they are not always the most effective option.

5.2 Comparison across three core legal tasks

In this section, we analyze how different modeling roles of LLMs contribute to various legal tasks. To ensure consistency and comparability across tasks with diverse evaluation criteria, we included studies reporting F-score–based metrics in this part of the quantitative analysis, as they are representative and provide a more comprehensive reflection of model performance by integrating both precision and recall. For each LLM role within a given task, we collect the relative changes in F-based scores and visualize their distributions using box plots. Each plot shows the median, interquartile range (IQR), and whisker bounds, enabling a robust and interpretable comparison of the effectiveness of different LLM roles.

Figure 10 presents the performance improvements of LLMs in two modeling roles, LLM Enhancement and Framework, across legal classification, retrieval, and generation tasks. Overall, both roles yield positive improvements, with distinct trends across task types. Generation tasks exhibit the most significant gains, with framework reshaping achieving a slightly higher and more stable median improvement compared to LLM enhancement, suggesting that incorporating LLMs into the overall workflow better leverages their generative capabilities. Classification tasks show relatively modest gains under both roles, reflecting the fact that these tasks often have strong baselines and limited room for improvement. Nevertheless, framework reshaping still provides a more consistent performance improvement. Retrieval tasks benefit from both roles with stable and moderate improvements, suggesting

that LLMs are particularly effective in improving semantic representations of inputs, which plays a central role in retrieval regardless of integration strategy. Overall, these results suggest that while LLM enhancement enables efficient performance gains, restructuring the framework to better align with LLM capabilities tends to yield more robust and consistent improvements across legal tasks.

6 Discussion

This section summarizes our overall findings across the previous sections, offering an overview of the current state of LLMs in LegalAI research from the perspectives of data, modeling, and evaluation, and outlines promising directions for future research.

From the data perspective, these LLMs have been actively applied to data augmentation and label generation, particularly in legal classification tasks where label structures are clearly defined and supervision is explicit. These tasks benefit from both input-level diversity and label-level alignment, allowing LLMs to generate meaningful variants and structured annotations. In contrast, legal retrieval and generation tasks remain underdeveloped in their use of LLMs for label-related processing. Applications involving label correction and reconstruction remain largely absent, and few studies construct complete pipelines that integrate different data processing stages with quality verification. Most interventions related to samples and labels are fragmented and task-specific, often lacking generalizability or integration into downstream workflows. This underdevelopment is primarily due to the lack of clearly defined label structures in retrieval and generation tasks. Their supervision signals are often implicit, relying on semantic similarity in retrieval tasks and open-text relevance in generation tasks, which makes it difficult to verify or standardize LLM-generated outputs within a consistent data processing pipeline. In addition to the above limitations, when LLMs generate training data, their outputs often reproduce and amplify pre-existing knowledge biases through a self-reinforcing loop of statistical learning. They tend to generalize along the dominant patterns embedded in the corpus, overproducing frequently occurring legal interpretations or majority viewpoints while underrepresenting minority or context-specific precedents, thereby resulting in a legal monoculture. Such tendencies reveal deeper epistemic biases across gender, cultural, socio-economic, and institutional dimensions (Dahl et al. 2024; Tan et al. 2023). However, current approaches to evaluating generated data rarely assess such biases explicitly, and most still rely on human evaluation, which itself is subject to cognitive and cultural biases. Moreover, current datasets are unevenly distributed across jurisdictions and languages. These patterns indicate that the potential of LLMs in data processing remains underexplored, particularly in complex or weakly supervised legal tasks where richer data semantics and multi-stage pipelines are needed to support more consistent and scalable model training. Future research could develop quantitative methods for measuring epistemic bias and tracing how such biases propagate through data generation pipelines, while moving beyond isolated improvements toward a more structured, multi-stage data optimization process that leverages LLMs to develop more unbiased, multilingual, and cross-jurisdictional legal datasets.

From the modeling perspective, the integration of LLMs into LegalAI systems demonstrates promising but uneven progress across tasks. Although both LLM enhancement and framework reshaping have been explored in classification, retrieval, and generation, most

approaches still focus on surface-level input augmentation, such as prompt-based tuning or semantic enrichment, with limited incorporation of LLMs into the modeling pipeline. While classification tasks, which benefit from clearer label structures and more mature modeling strategies, demonstrate a relatively stronger alignment between LLM capacities and model training objectives, retrieval and generation tasks have progressed more slowly, since integration is often less necessary in scenarios where LLMs alone are already sufficient or where traditional models outperform hybrid designs (Sie et al. 2024; Shui et al. 2023). This reflects a broader tendency to treat LLMs as external tools rather than structurally embedded components within modeling architectures, mainly because their long-chain reasoning capabilities have not yet been systematically enhanced or structurally modeled in current LegalAI research. In addition, current studies remain largely confined to single-language and single-jurisdiction settings, with limited cross-lingual or cross-jurisdictional comparisons. While LLMs have mitigated multilingual challenges to some extent, cross-jurisdictional research remains difficult due to differences in legal reasoning structures, data organization, and annotation standards. In the future, research could focus more on enhancing the structural reasoning capacity and internal interpretability of LLMs, enabling them to perform deeper and more autonomous legal reasoning. Furthermore, their multilingual and contextual reasoning capabilities could be leveraged to develop models that generalize across languages and jurisdictions.

From the evaluation perspective, the involvement of LLMs has introduced new evaluation scenarios across legal classification, retrieval, and generation tasks. These include not only traditional performance metrics but also emerging efforts to assess the quality and legal validity of model reasoning, particularly in generation tasks. Among the three, legal generation makes more extensive use of LLM-based scoring and human–LLM collaborative evaluation, enabling richer assessments of output quality. Output reconstruction, which could provide deeper insights into reasoning alignment and label consistency, remains rarely implemented. This is mainly because legal classification and retrieval tasks typically rely on fixed benchmark datasets, whose output formats are well-defined, while legal generation tasks produce more open-ended and semantically diverse outputs, where direct evaluation can be particularly challenging in certain scenarios, such as complex multi-dimensional assessments in court view generation and semantically dependent tasks like question generation or legal QA. However, as most current LegalAI models do not explicitly represent their reasoning processes or intermediate decision structures, output reconstruction remains difficult to implement, leading most studies to adopt LLM-based scoring and human–LLM collaborative evaluation instead. Overall, recent evaluation metrics and methods are often designed for specific emerging scenarios and remain largely grounded in traditional evaluation systems. As a result, current evaluation practices remain fragmented and task-specific, lacking unified frameworks for systematically assessing legal reasoning performance and normative alignment across increasingly diverse evaluation scenarios. Future work could develop scoring methods that leverage LLMs' internal reasoning, design task-specific reconstruction mechanisms, and explore human–LLM collaboration for assessing legal performance in emerging legal scenarios and reasoning-intensive contexts.

7 Conclusion

In this paper, we present a comprehensive survey of how LLMs are reshaping the research paradigm of LegalAI, especially in three main tasks. We propose a role-based schema that characterizes the involvement of LLMs across three core dimensions: data, modeling, and evaluation. Based on this schema, we systematically review existing work on legal classification, retrieval, and generation. Besides, we quantitatively compare LLM effectiveness across roles and tasks. Our analysis reveals that LLMs are no longer limited to performing tasks, but instead restructure the design and resolution strategies of legal tasks. This transformation highlights both the strengths and limitations of different LLM roles and offers a conceptual perspective for understanding the paradigm shift in LegalAI research. In the last part, we discuss promising directions for future research.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

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Consent for publication Not applicable.

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